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EDUCATION TO INNOVATION

DSM5012C

Data Analytics Using Tools

Unit No: 3 Data Analytics using EXCEL

SEMESTER: 2

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Introduction to Data Analytics in Excel

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What is Data Analytics?

- Data Analytics is the **process of examining raw data** to extract useful information.
- It helps in **finding patterns, trends, and insights** from data.
- Used to support **decision-making** in organizations.

👉 **Example:**
Analyzing sales data to find which product sells the most.

Key Objectives of Data Analytics

- Convert raw data into meaningful insights
- Improve decision-making
- Identify trends and patterns
- Increase business efficiency
- Predict future outcomes

What is Excel in Data Analytics?

- **Microsoft Excel** is a spreadsheet tool used for:
  - Data storage
  - Data analysis
  - Data visualization
- It is one of the **most widely used tools** for beginners in analytics

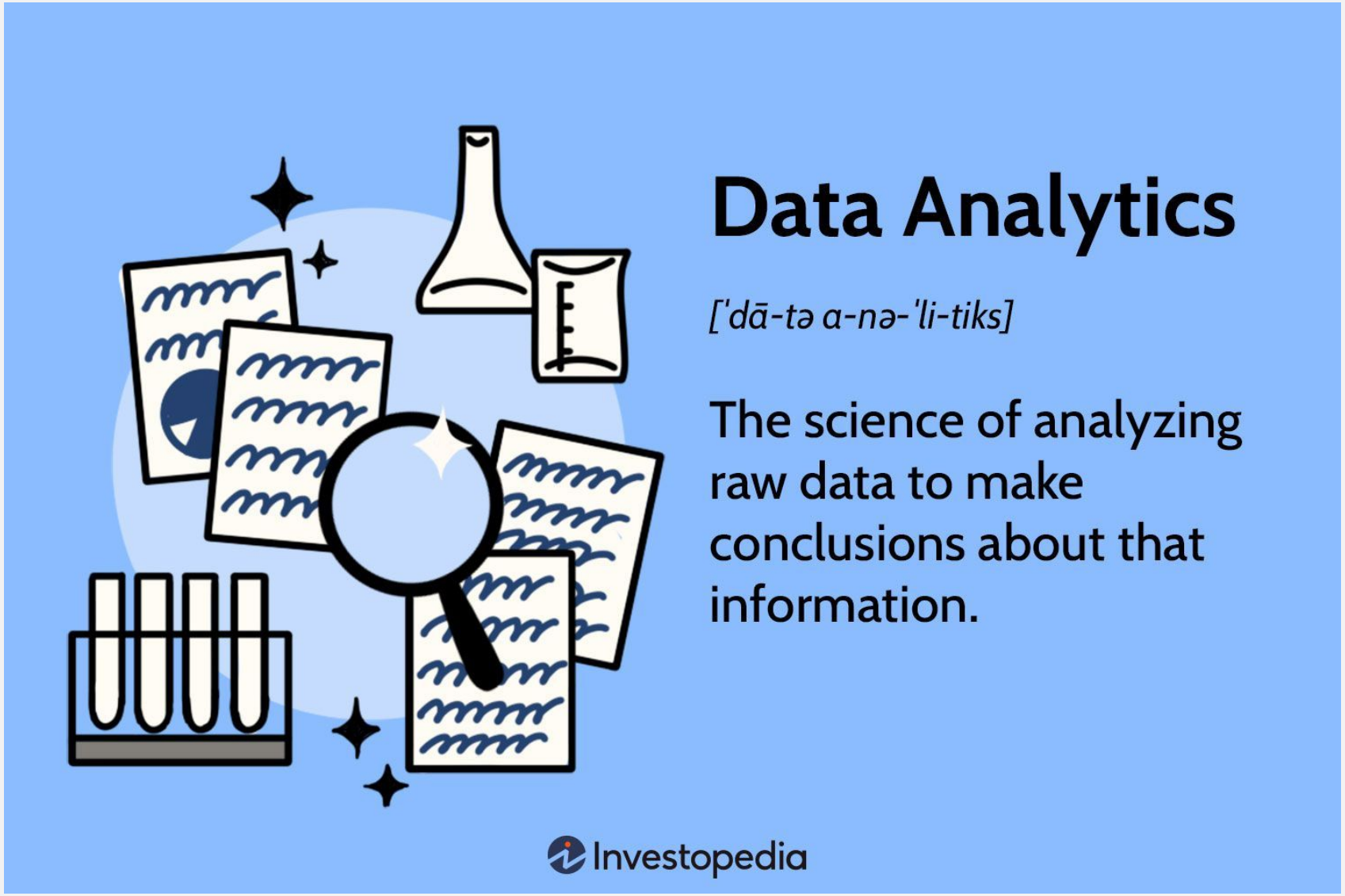
Why Use Excel for Data Analytics?

- Easy to learn and use
- No programming required
- Built-in formulas and functions
- Supports charts and dashboards
- Suitable for small to medium datasets

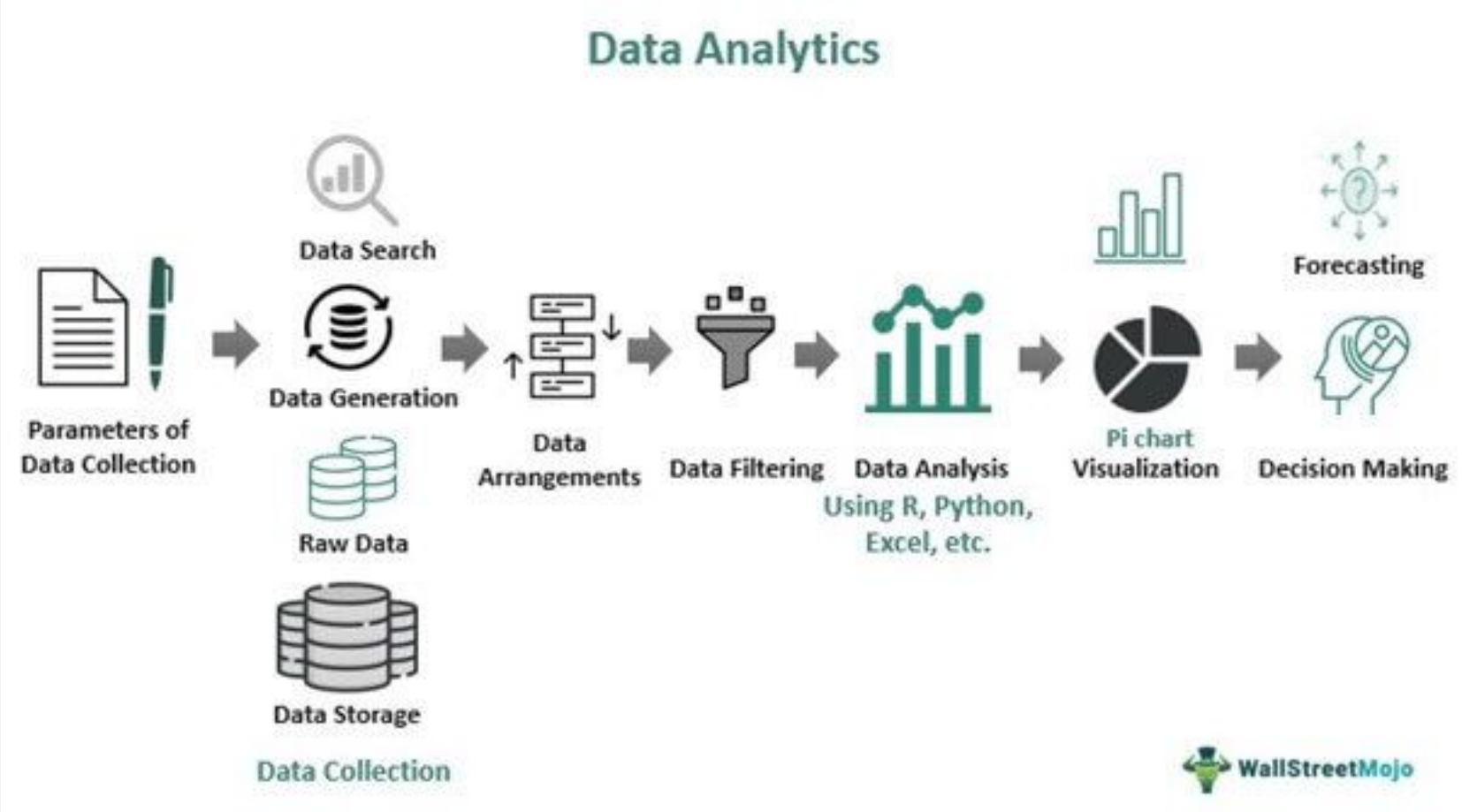
Features of Excel for Data Analytics

- Data organization using rows & columns
- Functions like:
  - AVERAGE()
  - SUM()
  - IF()
- Data visualization (charts, graphs)
- Pivot Tables for summarization
- Data Analysis ToolPak for statistical analysis

Introduction to Data Analytics in Excel



Introduction to Data Analytics in Excel



Introduction to Data Analytics in Excel

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Types of Data Analytics (Brief Overview)

1. **Descriptive Analytics**
   - What happened?
   - Example: Monthly sales report
2. **Diagnostic Analytics**
   - Why did it happen?
   - Example: Drop in sales analysis
3. **Predictive Analytics**
   - What will happen?
   - Example: Sales forecasting
4. **Prescriptive Analytics**
   - What should be done?
   - Example: Business strategy suggestions

Real-Life Applications

- Business: Sales analysis, profit tracking
- Finance: Budget planning
- Education: Student performance analysis
- Healthcare: Patient data analysis

Advantages of Using Excel

- Cost-effective
- User-friendly interface
- Quick data analysis
- Widely available
- Good for beginners

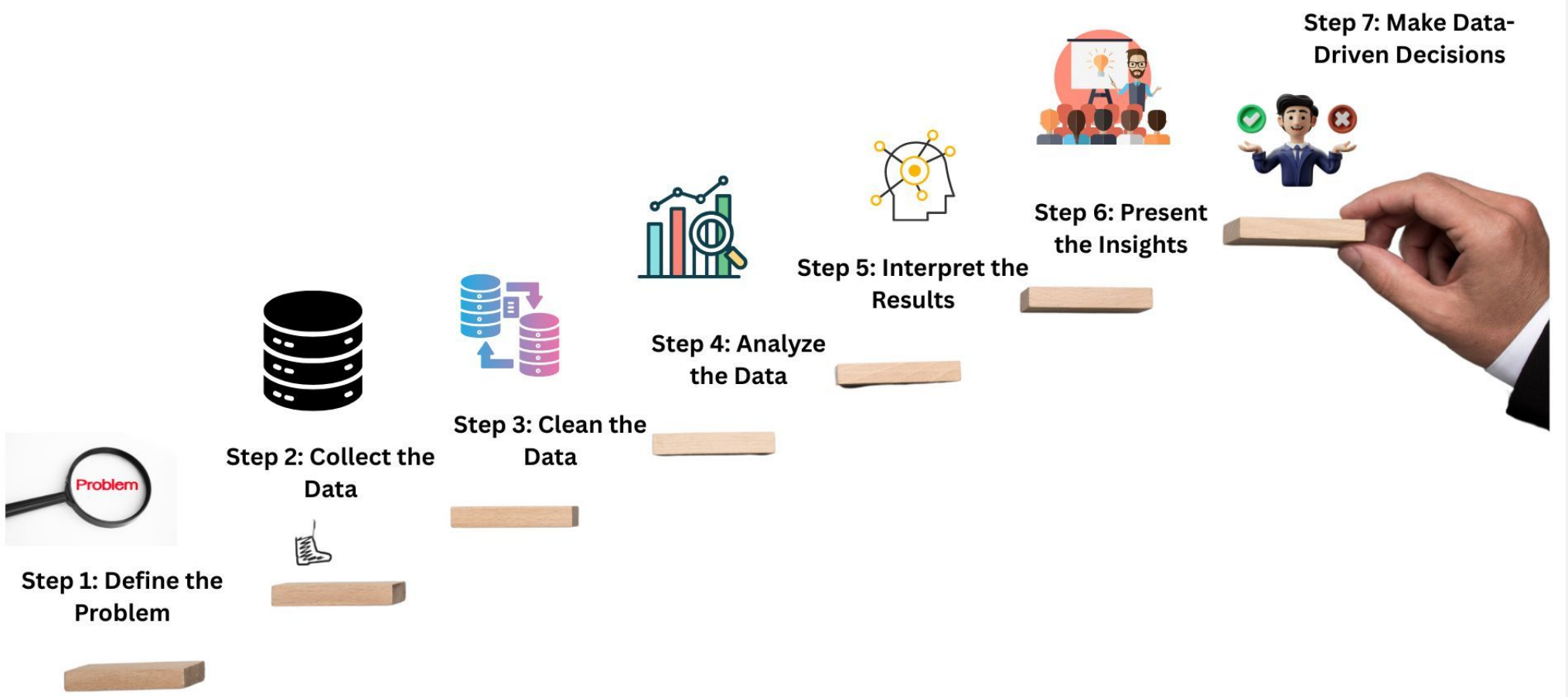
Limitations of Excel

- Not suitable for very large datasets
- Limited automation compared to advanced tools
- Manual errors possible
- Less powerful than tools like Python or R

Conclusion

- Data Analytics helps in making data-driven decisions
- Excel is a powerful starting tool for analytics
- It provides essential features for:
  - Data cleaning
  - Analysis
  - Visualization

Introduction to Data Analytics in Excel



7 Steps of Data Analysis for Predictive Analytics & Business Insights

VISTA ACADEMY

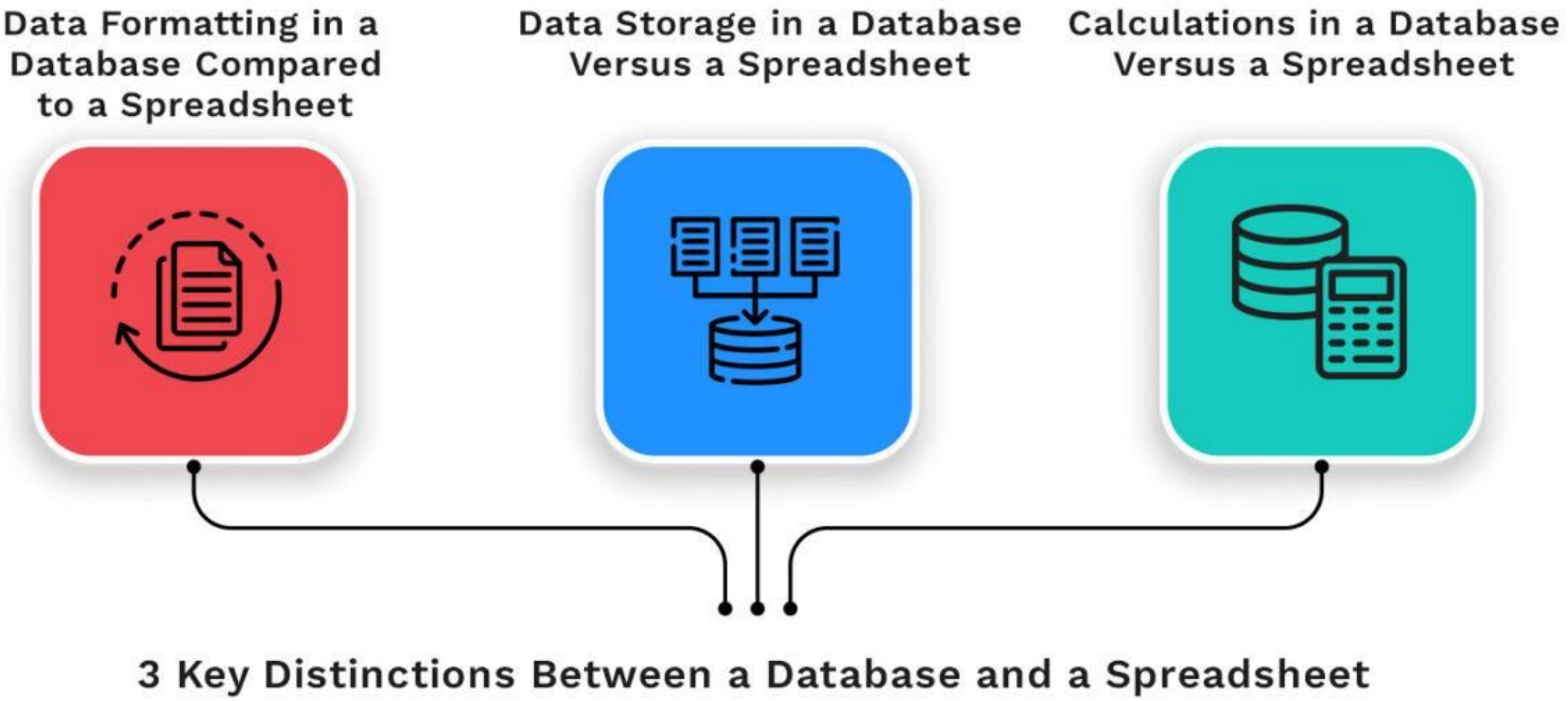
Role of Spreadsheet Tools

Role	Description	Excel Features Used	Example
Data Storage & Organization	Stores data in structured format using rows and columns	Worksheets, Tables	Student database, Sales records
Data Cleaning	Improves data quality by removing errors and inconsistencies	Remove Duplicates, TRIM(), ISBLANK()	Removing duplicate entries
Data Analysis	Performs calculations and identifies patterns	SUM(), AVERAGE(), IF()	Total sales calculation
Data Visualization	Represents data using charts and graphs	Bar Chart, Line Chart, Pie Chart	Sales trend graph
Statistical Analysis	Applies statistical methods to analyze data	MEDIAN(), MODE(), STDEV()	Finding average marks
Automation	Reduces manual work using formulas and macros	Formulas, Macros	Auto calculation of results
Reporting	Creates summaries and reports for decision-making	Pivot Tables, Charts	Monthly report
Dashboard Creation	Builds interactive visual dashboards	Pivot Charts, Slicers	Business performance dashboard
Data Filtering & Sorting	Helps in arranging and filtering data	Sort, Filter options	Top 10 products
Data Transformation	Converts data into useful format	CONCAT(), Text to Columns	Combining first & last name

Role of Spreadsheet Tools

- Maintain a Backup of Raw Data
- Use Excel Tables for Structured Data Management
- Implement Data Validation from the Start
- Leverage Conditional Formatting to Spot Anomalies
- Handle Missing Data Effectively
- Standardize Text and Date Formatting
- Regularly Audit and Validate Data
- Document Your Cleaning Process

Role of Spreadsheet Tools



Excel Interface & Data Analysis ToolPak

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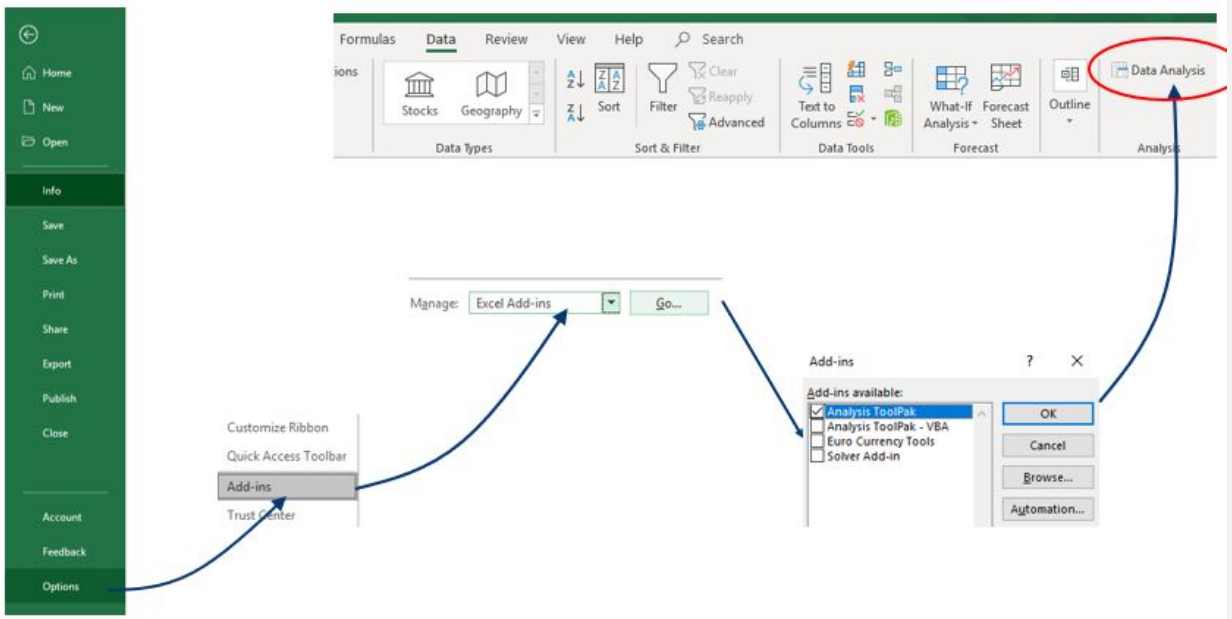
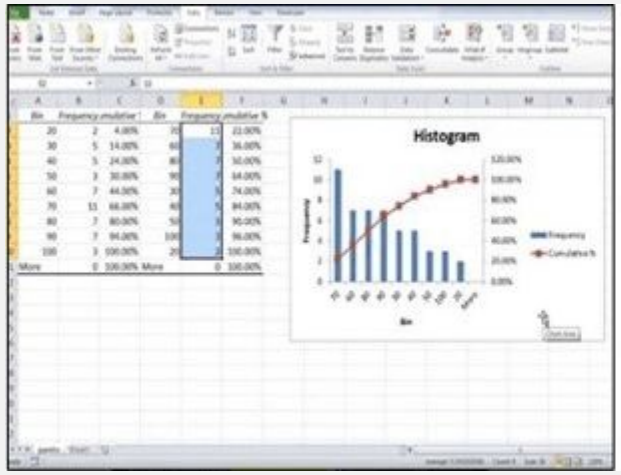
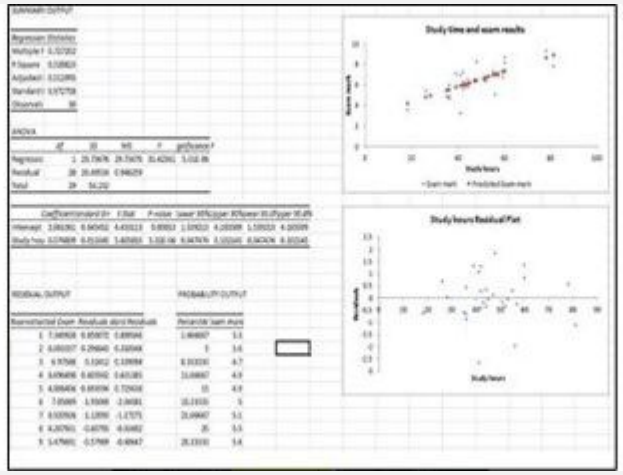
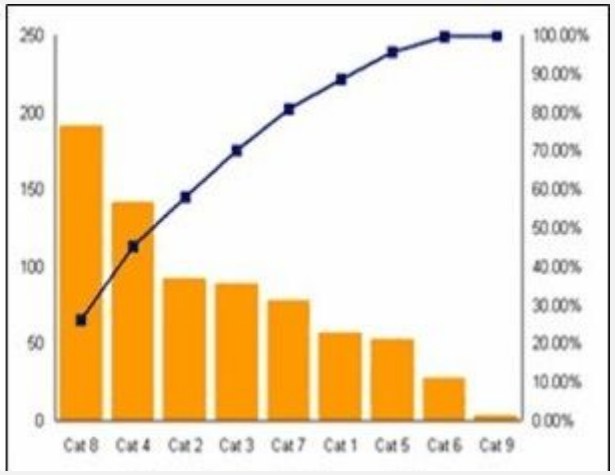
No.	Topic	Description	Example / Use
1	Excel Interface	The working environment of Microsoft Excel where users interact with data	Opening Excel window
2	Ribbon	Top menu containing tabs like Home, Insert, Data, etc.	Access formatting, charts
3	Worksheet	Grid of rows and columns where data is entered	Data entry in cells
4	Rows & Columns	Rows (horizontal), Columns (vertical) forming cells	Row 1, Column A
5	Cell	Intersection of row and column where data is stored	A1, B2
6	Formula Bar	Area to enter or edit formulas and data	Writing =SUM(A1:A10)
7	Name Box	Shows address of selected cell	Displays A1
8	Data Analysis ToolPak	Add-in providing advanced statistical tools	Regression, correlation
9	ToolPak Installation	Enable from File → Options → Add-ins → Analysis ToolPak	Activate features
10	ToolPak Features	Includes Descriptive Stats, Regression, ANOVA, Moving Average	Data analysis tasks
11	Use of ToolPak	Used for statistical analysis and forecasting	Business analysis

Excel Interface & Data Analysis ToolPak





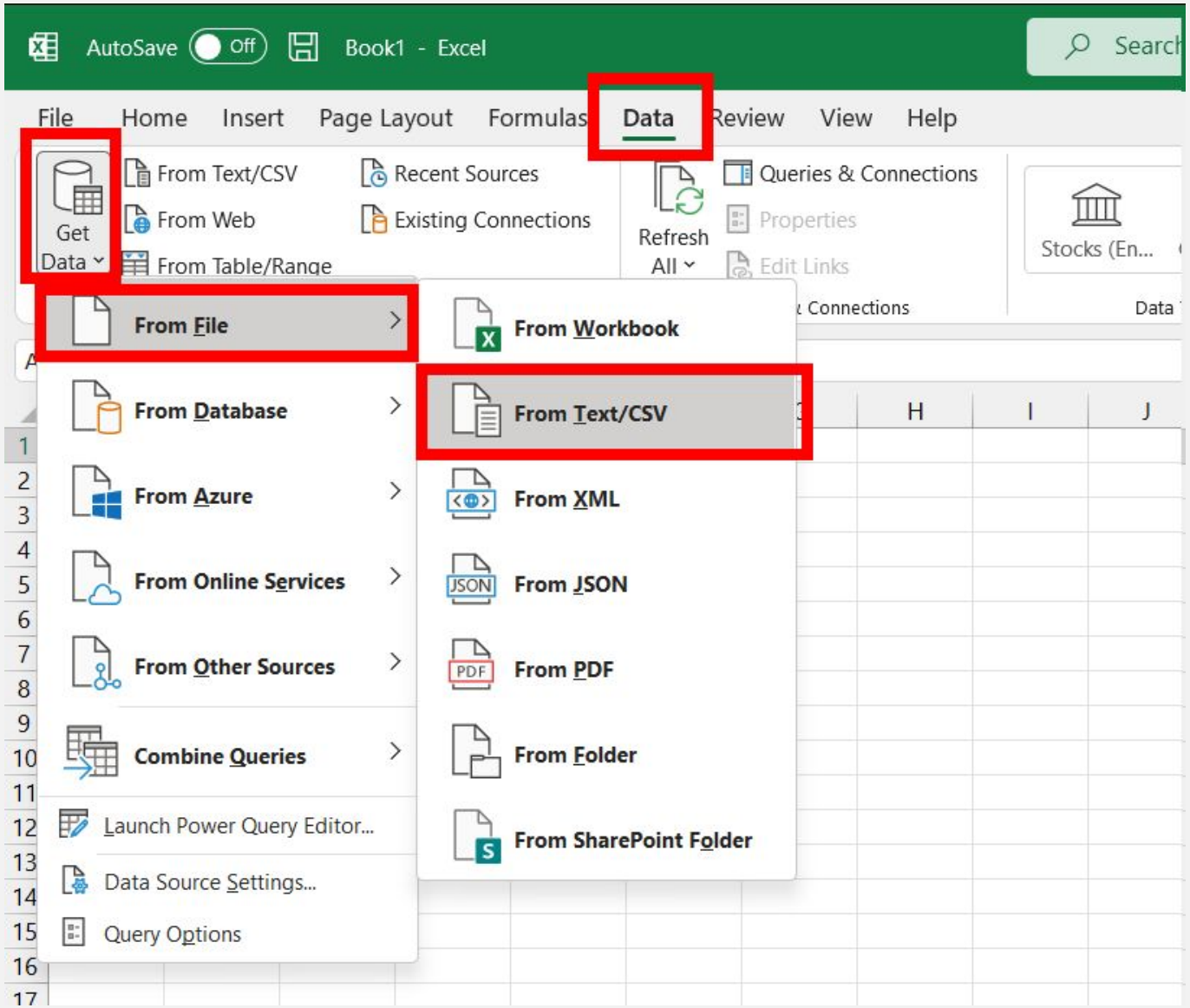


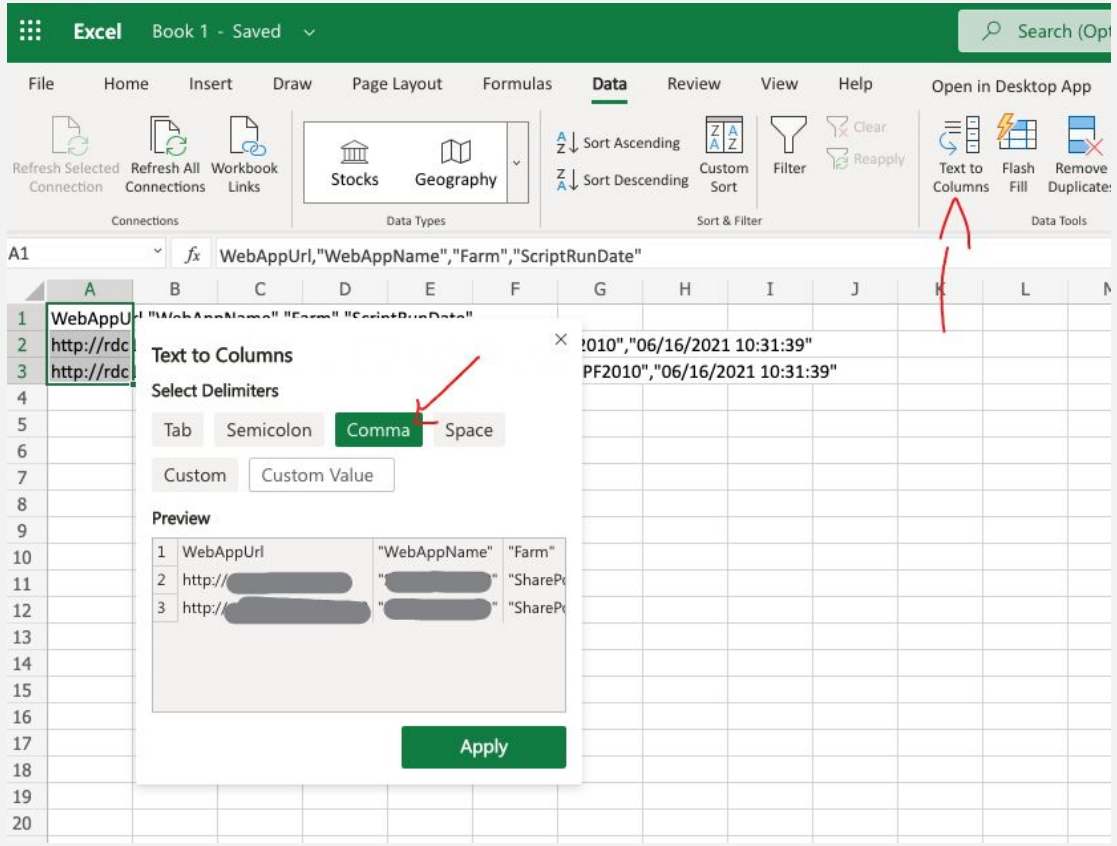


Importing Data into Excel

No.	Topic	Description	Example / Use
1	Importing Data	Process of bringing external data into Excel for analysis	Loading CSV file
2	Data Sources	Various sources from which data can be imported	CSV, Text, Database, Web
3	CSV Files	Comma Separated Values file format	Sales data.csv
4	Text Files	Plain text files (.txt) with data	Tab-separated data
5	Excel Files	Import data from another Excel workbook	Copy sheets/data
6	Web Data	Import data from websites using URL	Stock prices
7	Get Data Tool	Excel feature to import data from multiple sources	Data Tab → Get Data
8	Power Query	Advanced tool for data import and transformation	Data cleaning during import
9	Data Refresh	Update imported data automatically	Refresh stock data
10	Data Transformation	Modify data during import	Filter, split columns

Importing Data into Excel

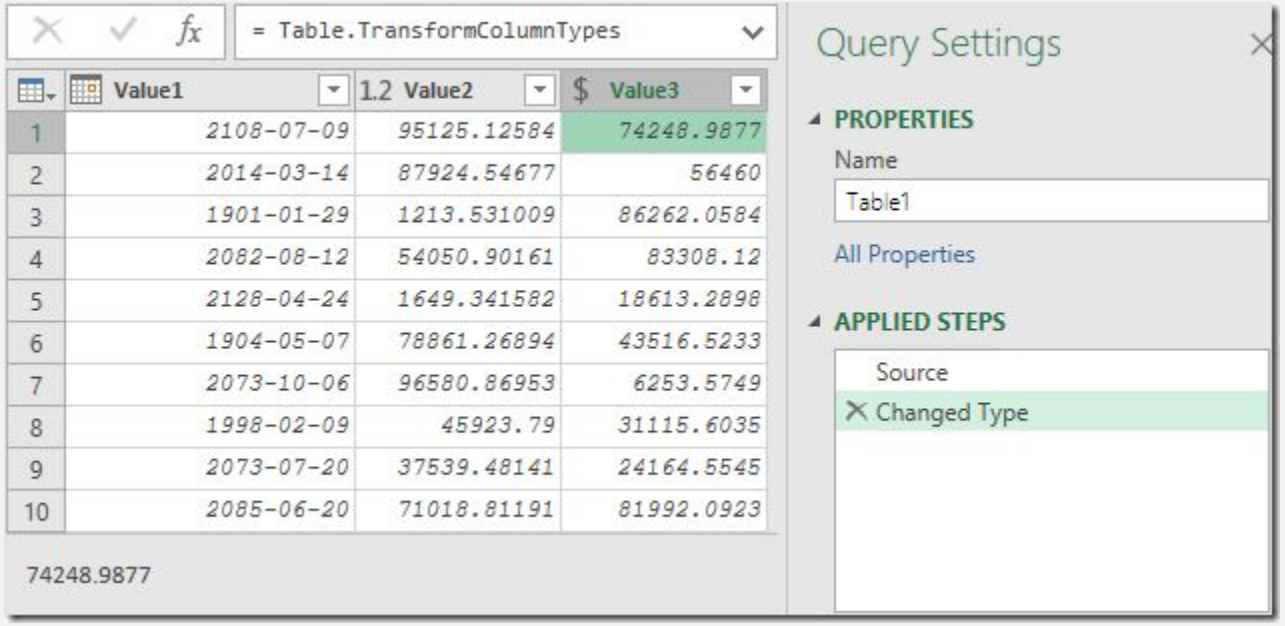


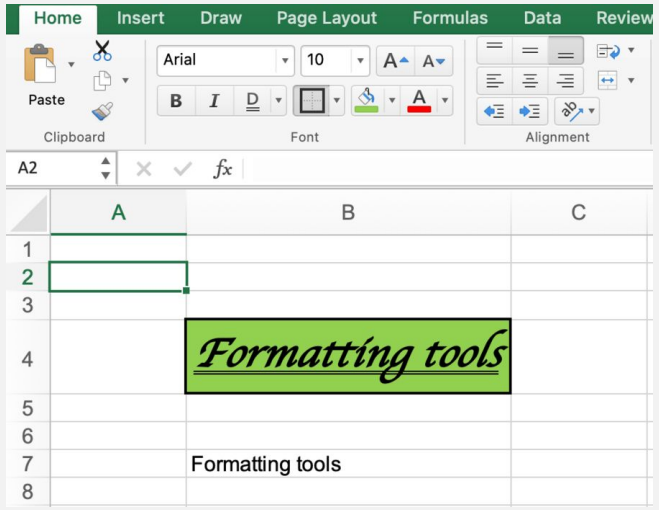


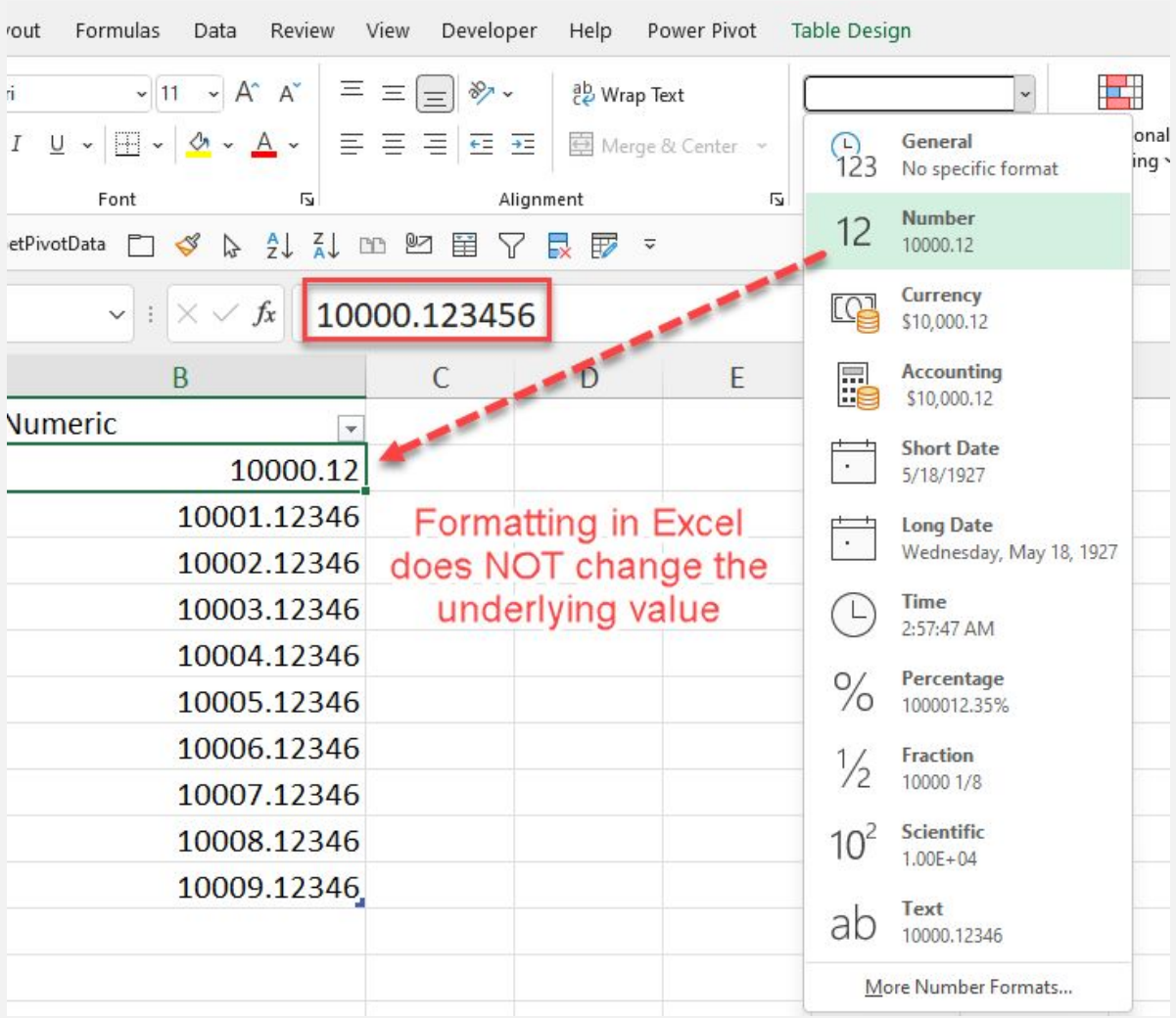
Data Types & Formats

No.	Data Type / Format	Description	Example
1	Numeric	Numbers used for calculations	100, 25.5
2	Text	Alphanumeric data, not used in calculations	Name, City
3	Date & Time	Stores dates and time values	01-01-2026, 10:30 AM
4	Currency	Represents monetary values	₹5000
5	Percentage	Represents values in percentage form	75%
6	Logical (Boolean)	TRUE or FALSE values	TRUE, FALSE
7	General Format	Default format in Excel	Mixed data
8	Number Format	Controls decimal places, separators	1,000.50
9	Custom Format	User-defined formatting	Phone number format
10	Scientific Format	Displays large numbers in exponential form	1.2E+05

Data Types & Formats



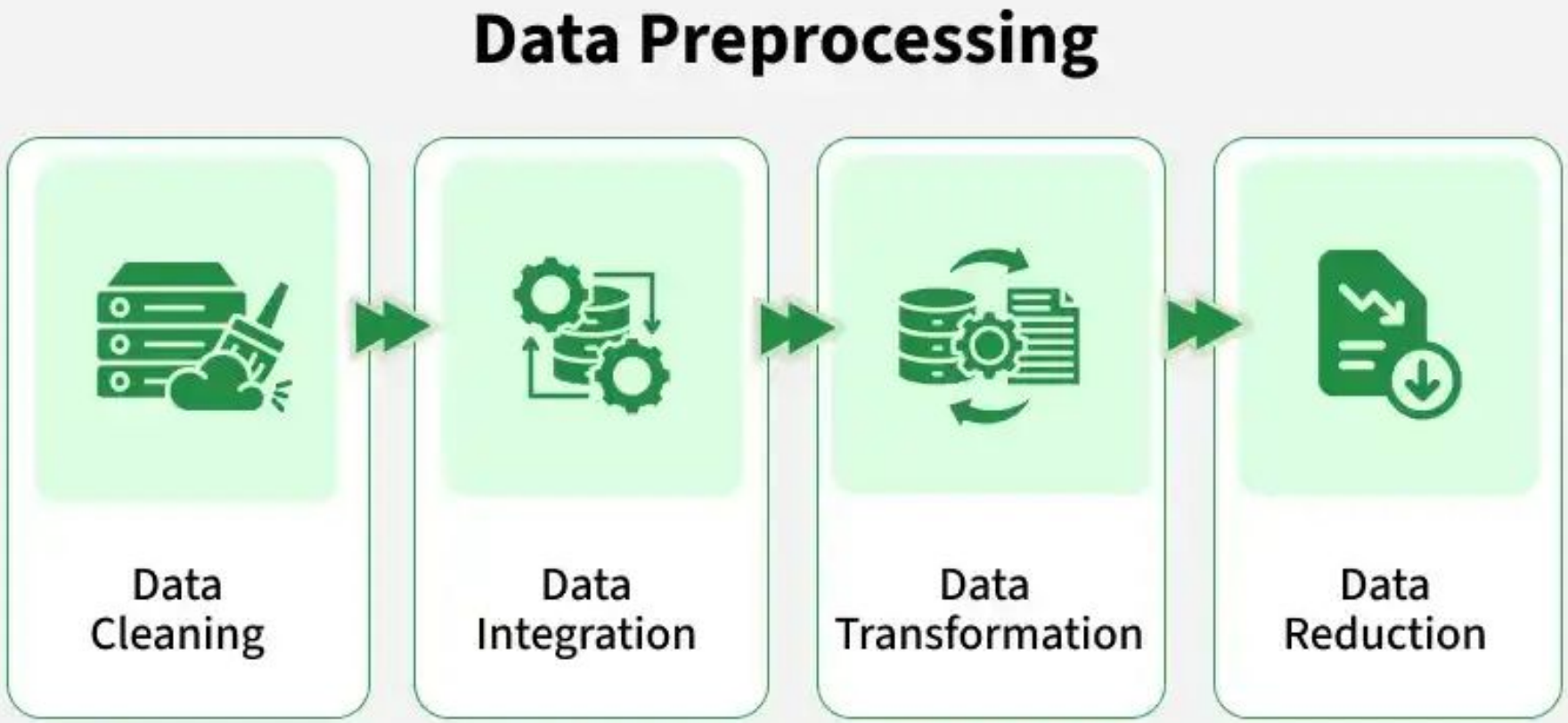


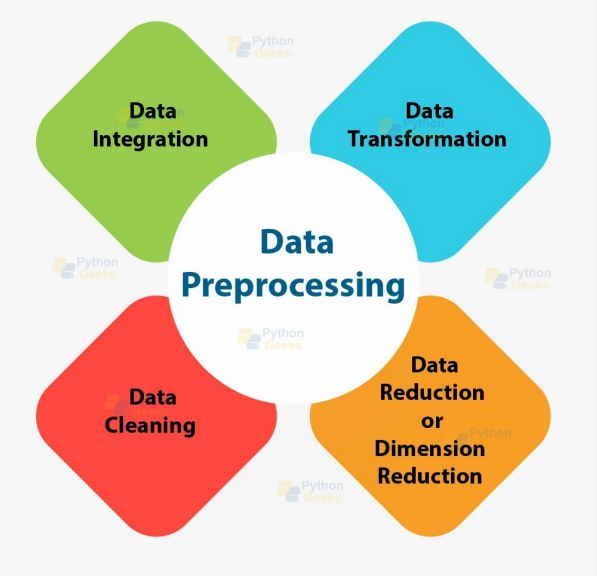


Data Preprocessing Techniques

No.	Technique	Description	Example
1	Data Preprocessing	Process of preparing raw data for analysis	Cleaning dataset
2	Sorting	Arranging data in ascending/descending order	Sort marks
3	Filtering	Displaying selected data based on conditions	Filter students with marks > 80
4	Removing Duplicates	Eliminating repeated data entries	Remove duplicate names
5	Handling Missing Values	Managing blank or null values	Replace blanks with average
6	Data Validation	Restricting input values	Only numbers allowed
7	Data Normalization	Scaling data into a standard range	Values between 0 and 1
8	Data Formatting	Ensuring consistent format	Date format standardization
9	Text to Columns	Splitting data into multiple columns	Split full name
10	Find & Replace	Modify specific values	Replace “N/A” with blank

Data Preprocessing Techniques





Data Cleaning using Excel Functions

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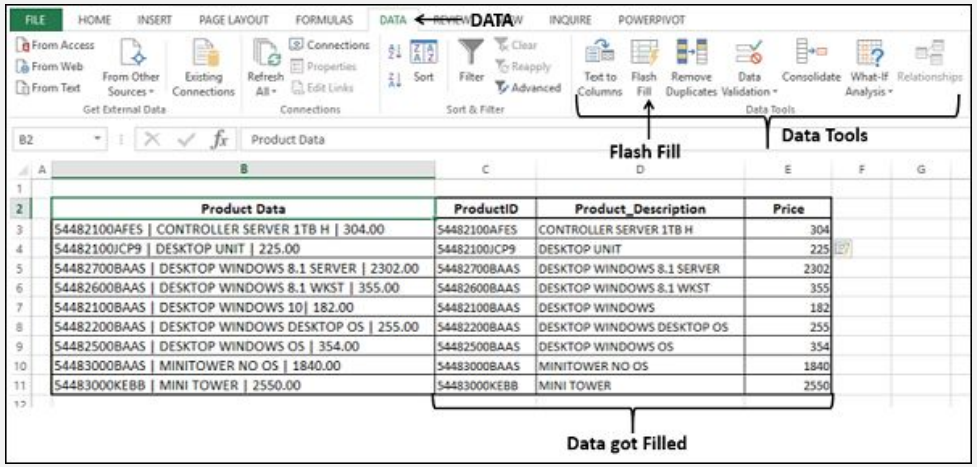
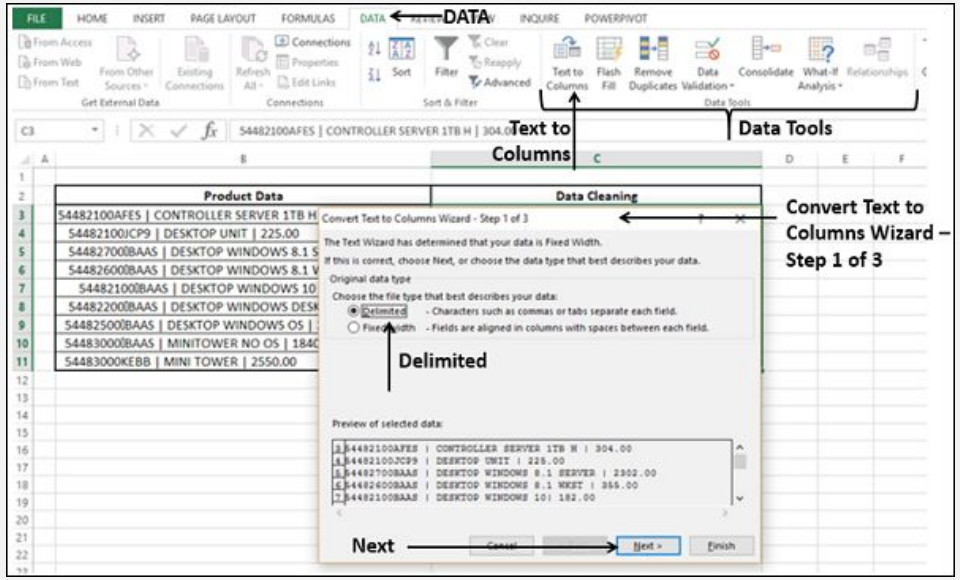
No.	Function / Technique	Description	Example
1	Data Cleaning	Process of correcting errors and improving data quality	Fixing messy dataset
2	TRIM()	Removes extra spaces from text	" Hello " → "Hello"
3	UPPER()	Converts text to uppercase	"excel" → "EXCEL"
4	LOWER()	Converts text to lowercase	"DATA" → "data"
5	PROPER()	Capitalizes first letter of each word	"john doe" → "John Doe"
6	IF()	Performs conditional cleaning	Replace incorrect values
7	ISBLANK()	Checks if a cell is empty	TRUE if blank
8	VLOOKUP() / XLOOKUP()	Finds and replaces values from another table	Match ID with name
9	CONCAT() / CONCATENATE()	Combines multiple text values	First + Last name
10	SUBSTITUTE()	Replaces specific text	Replace "N/A" with "0"

Data Cleaning using Excel Functions

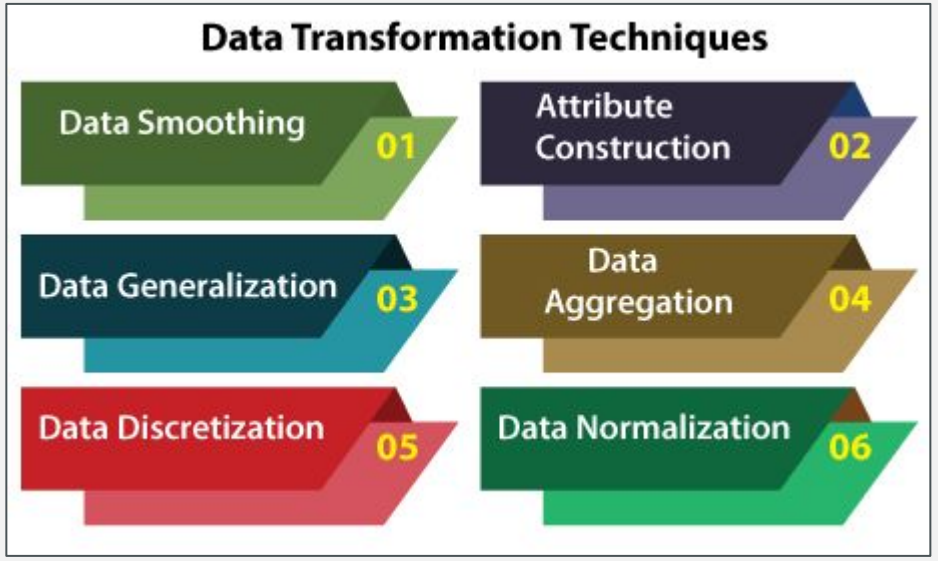


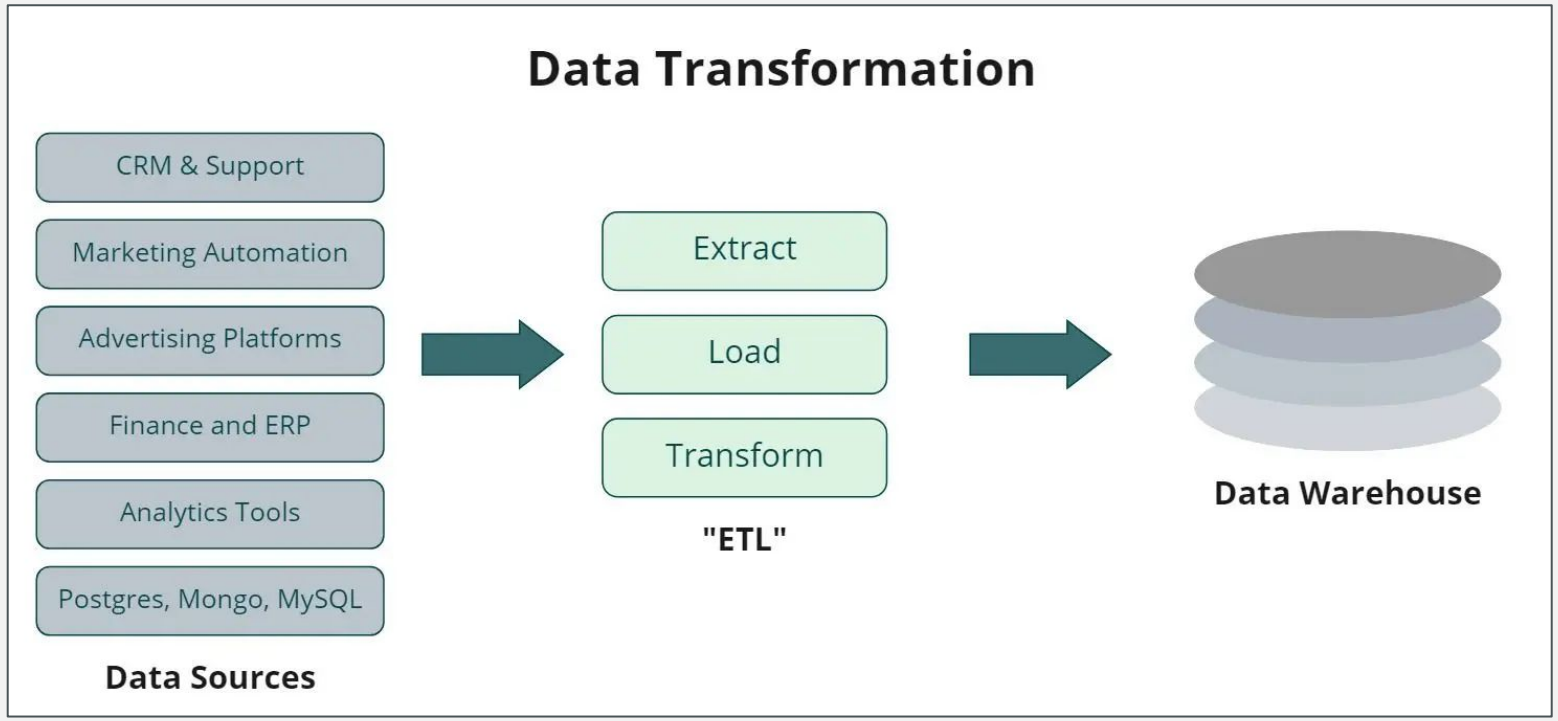


Data Transformation Techniques

No.	Technique / Function	Description	Example
1	Data Transformation	Process of converting data into a suitable format for analysis	Raw → structured data
2	CONCAT()	Combines multiple text values	First + Last name
3	TEXT()	Converts numbers into formatted text	Date to text format
4	VALUE()	Converts text into numeric value	"100" → 100
5	LEFT(), RIGHT(), MID()	Extracts specific part of text	Extract first name
6	Text to Columns	Splits data into multiple columns	Split full name
7	Flash Fill	Automatically fills data patterns	Format phone numbers
8	Pivot Table	Summarizes and transforms data	Sales summary
9	Find & Replace	Modify specific values	Replace "Male" with "M"
10	Data Formatting	Change data structure/appearance	Date format change

Data Transformation Techniques

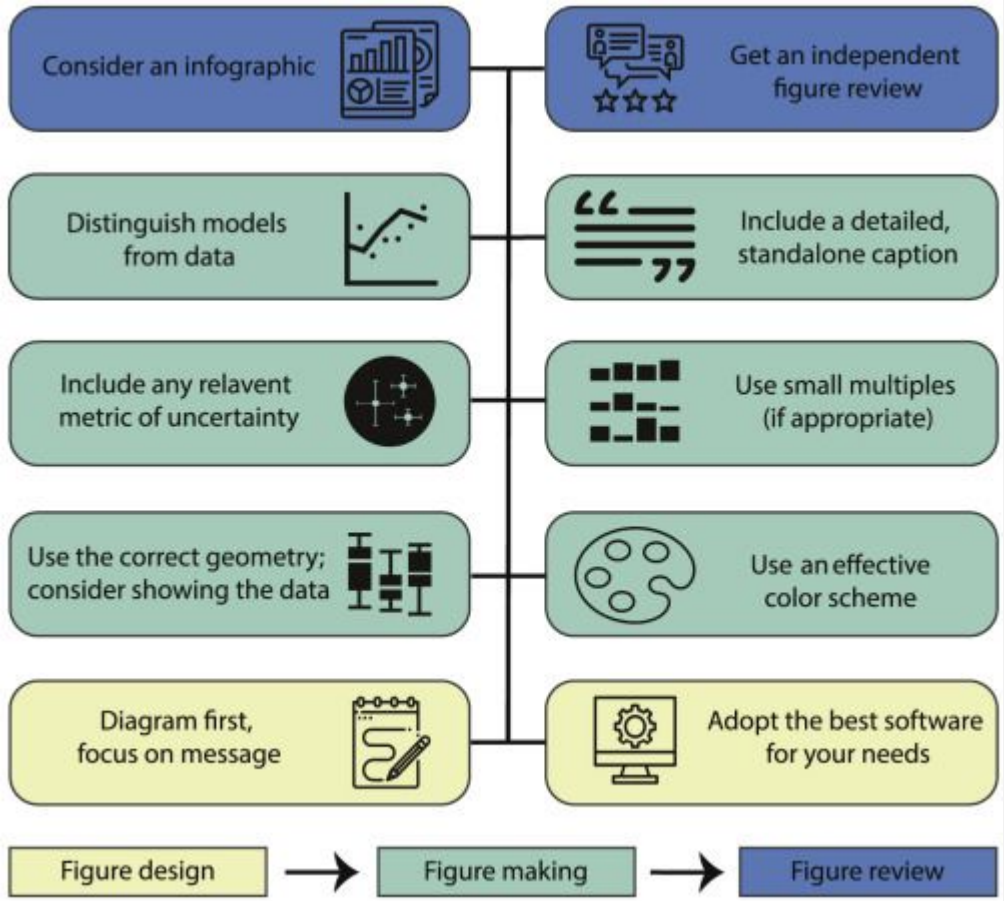




Data Visualization Principles

No.	Principle	Description	Example
1	Data Visualization	Representation of data using charts and graphs	Sales chart
2	Simplicity	Keep charts clean and easy to understand	Simple bar chart
3	Clarity	Clearly present information with labels and titles	Labeled graph
4	Accuracy	Represent data correctly without distortion	Correct scale graph
5	Appropriate Chart Selection	Choose correct chart based on data type	Line for trends
6	Consistency	Maintain uniform colors, fonts, and styles	Same color scheme
7	Highlight Key Insights	Emphasize important data points	Highlight highest value
8	Avoid Clutter	Remove unnecessary elements	No extra gridlines
9	Use of Colors	Use colors effectively to differentiate data	Multi-color chart
10	Proper Labels & Titles	Add axis labels, legends, and titles	Chart with labels

Data Visualization Principles



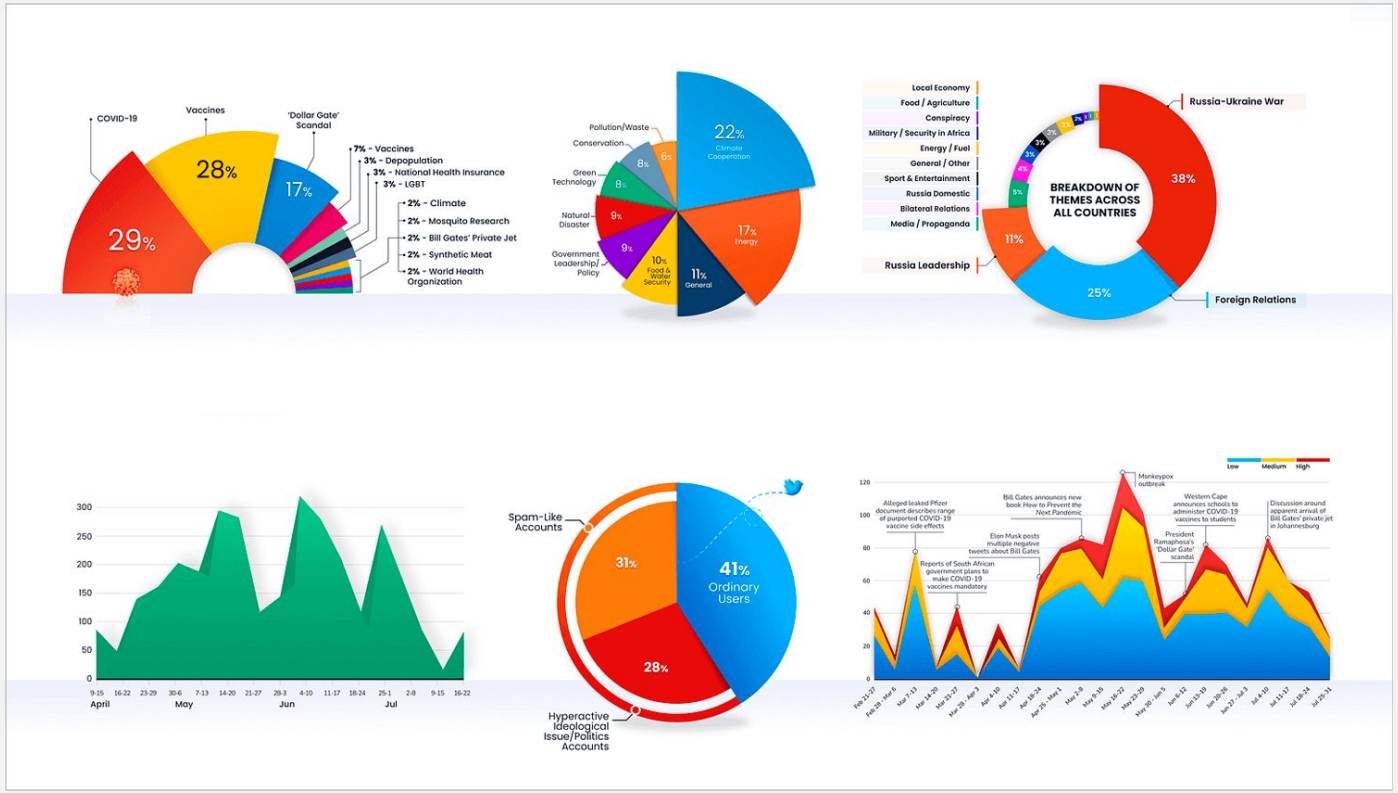
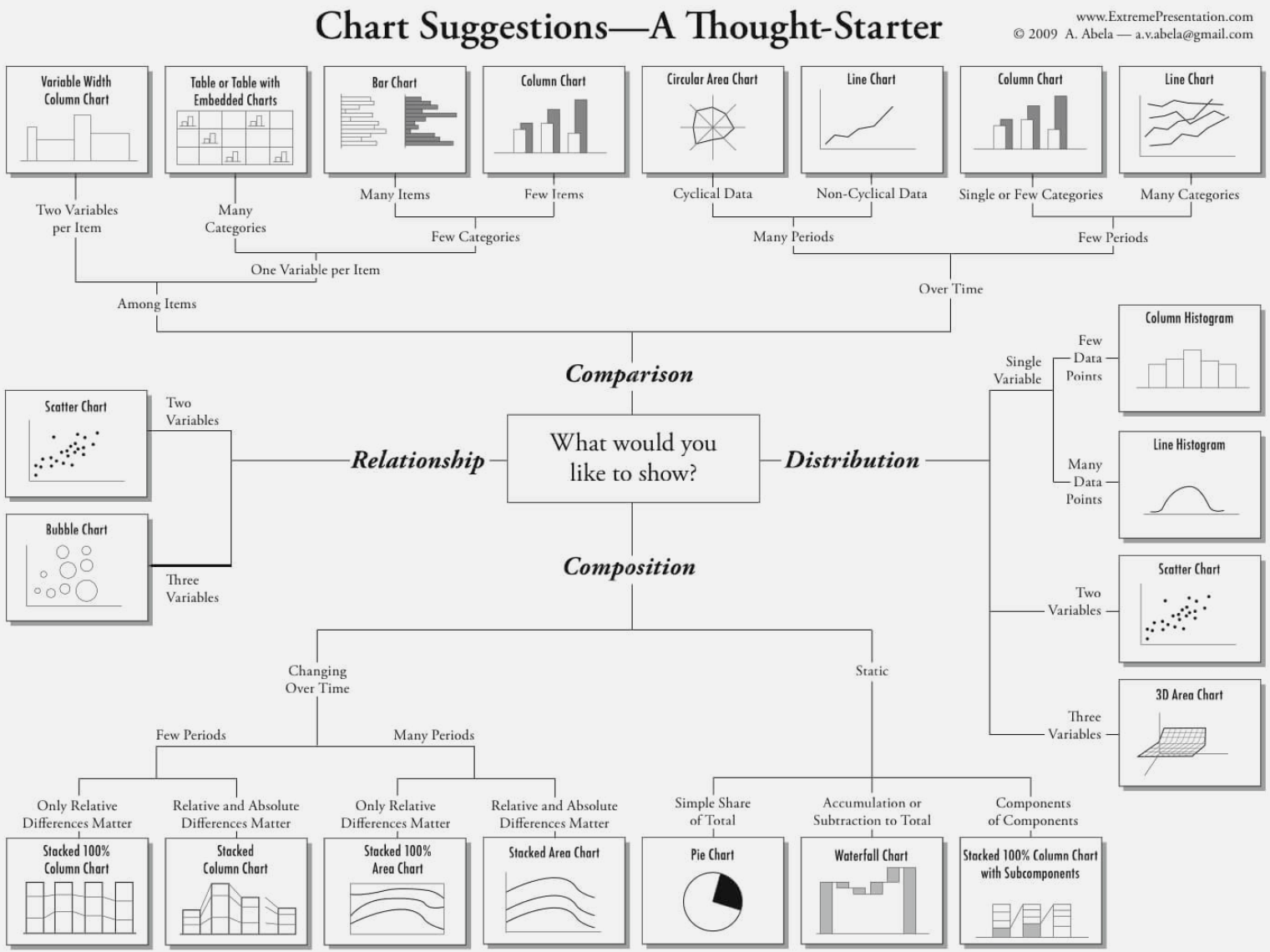


Chart Types & Selection

No.	Chart Type	Description	Best Use / Example
1	Bar Chart	Represents data using horizontal/vertical bars	Compare sales of products
2	Column Chart	Vertical bars for comparison	Monthly revenue comparison
3	Line Chart	Shows trends over time	Sales growth over months
4	Pie Chart	Shows proportion of parts to whole	Market share
5	Histogram	Displays frequency distribution	Exam score distribution
6	Scatter Plot	Shows relationship between variables	Height vs Weight
7	Area Chart	Similar to line chart with filled area	Cumulative data
8	Doughnut Chart	Variation of pie chart	Multi-category proportion
9	Combo Chart	Combines two chart types	Sales & profit comparison
10	Box & Whisker	Shows data distribution and outliers	Statistical analysis

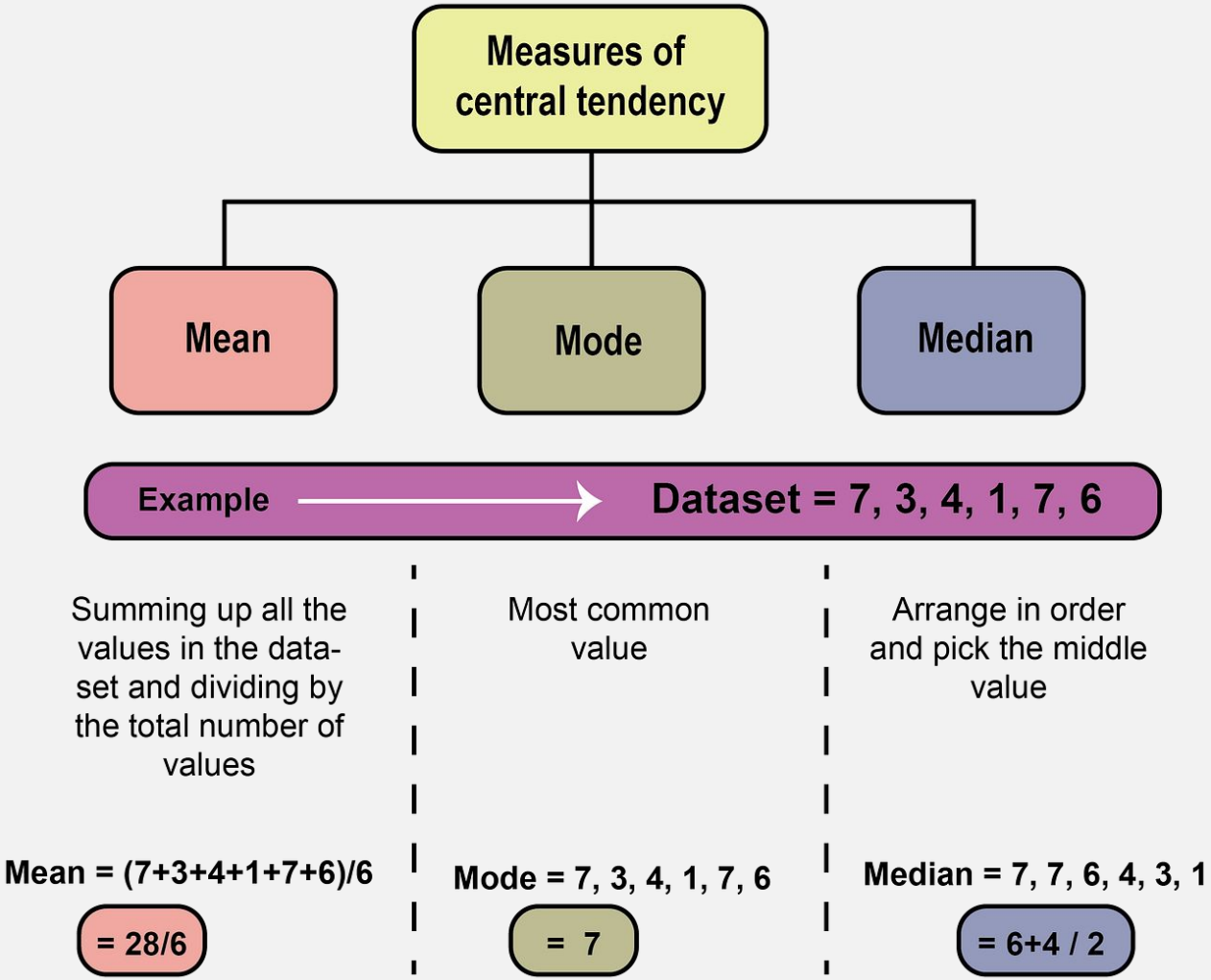
Chart Types & Selection



Measures of Central Tendency

Measure	Definition	Formula (Math)	Excel Function	Example Data	Excel Formula Example	Result
Mean (Average)	Sum of all values divided by number of values	Mean = ΣX / N	=AVERAGE(range)	10, 20, 30, 40, 50	=AVERAGE(A1:A5)	30
Median	Middle value when data is sorted	If odd → middle valueIf even → avg of 2 middle values	=MEDIAN(range)	10, 20, 30, 40, 50	=MEDIAN(A1:A5)	30
Mode	Most frequently occurring value	Value with highest frequency	=MODE.SNGL(range)	10, 20, 20, 30, 40	=MODE.SNGL(A1:A5)	20
Weighted Mean	Average considering weights	Σ(W×X) / ΣW	=SUMPRODUCT(values,weights)/SUM(weights)	Values: 10,20,30Weights: 1,2,3	=SUMPRODUCT(A1:A3,B1:B3)/SUM(B1:B3)	23.33
Geometric Mean	Product of values raised to power 1/n	$(X_1 \times X_2 \times \ldots \times Xn)^{(1/n)}$	=GEOMEAN(range)	2, 4, 8	=GEOMEAN(A1:A3)	4
Harmonic Mean	Reciprocal of average of reciprocals	n / Σ(1/X)	=HARMEAN(range)	2, 4, 8	=HARMEAN(A1:A3)	3.43

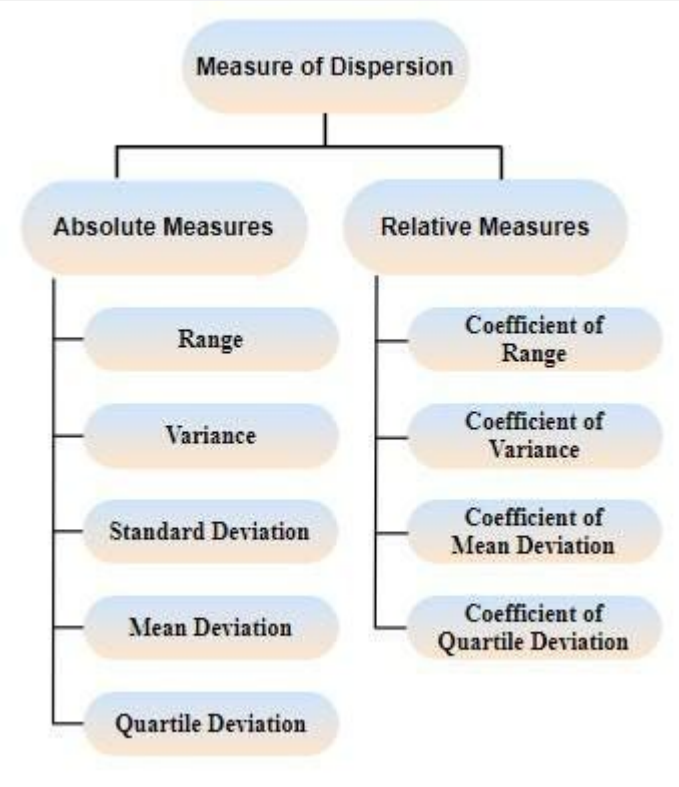
Measures of Central Tendency

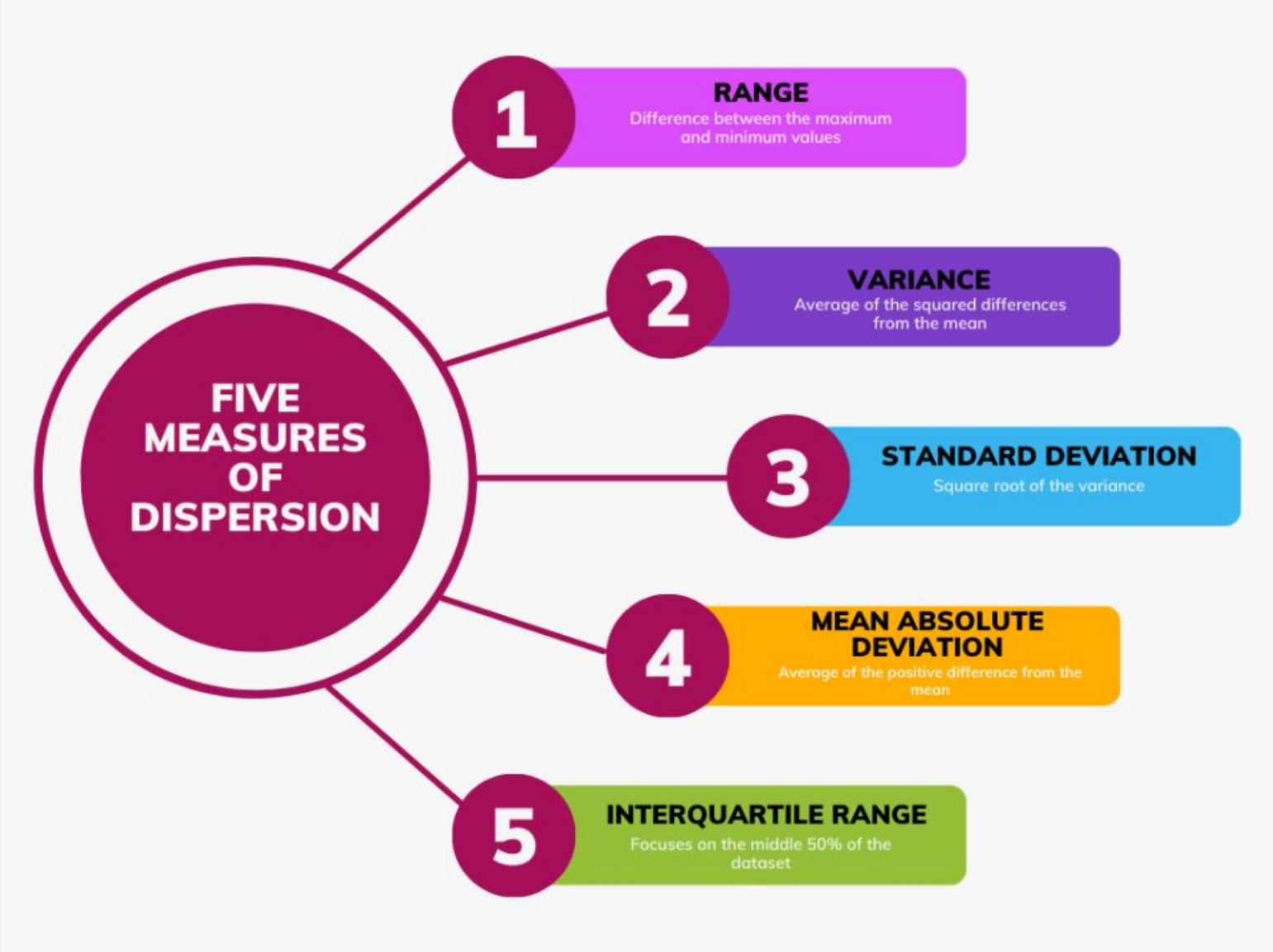


Measures of Dispersion

Measure	Definition	Formula (Math)	Excel Function	Example Data	Excel Formula Example	Result
Range	Difference between maximum and minimum values	Range = Max − Min	=MAX(range)-MIN(range)	10, 20, 30, 40, 50	=MAX(A1:A5)-MIN(A1:A5)	40
Variance (Population)	Average of squared deviations from mean	$\sigma^2 = \Sigma(X-\mu)^2 / N$	=VAR.P(range)	10, 20, 30, 40, 50	=VAR.P(A1:A5)	200
Variance (Sample)	Variance for sample data	$s^2 = \Sigma(X-\bar{x})^2 / (n-1)$	=VAR.S(range)	10, 20, 30, 40, 50	=VAR.S(A1:A5)	250
Standard Deviation (Population)	Square root of variance	$\sigma = \sqrt{\text{Variance}}$	=STDEV.P(range)	10, 20, 30, 40, 50	=STDEV.P(A1:A5)	14.14
Standard Deviation (Sample)	Spread for sample data	$s = \sqrt{\text{Variance}}$	=STDEV.S(range)	10, 20, 30, 40, 50	=STDEV.S(A1:A5)	15.81
Quartiles	Divide data into 4 equal parts	Q1, Q2, Q3	=QUARTILE.INC(range,quart)	10, 20, 30, 40, 50	=QUARTILE.INC(A1:A5,1)	20
Interquartile Range (IQR)	Spread of middle 50% data	IQR = Q3 − Q1	Use Quartile formula	10, 20, 30, 40, 50	=QUARTILE.INC(A1:A5,3)-QUARTILE.INC(A1:A5,1)	20
Mean Absolute Deviation (MAD)	Avg of absolute differences from mean	Σ	X−Mean	/ N	=AVEDEV(range)	10, 20, 30, 40, 50

Measures of Dispersion





Correlation Analysis

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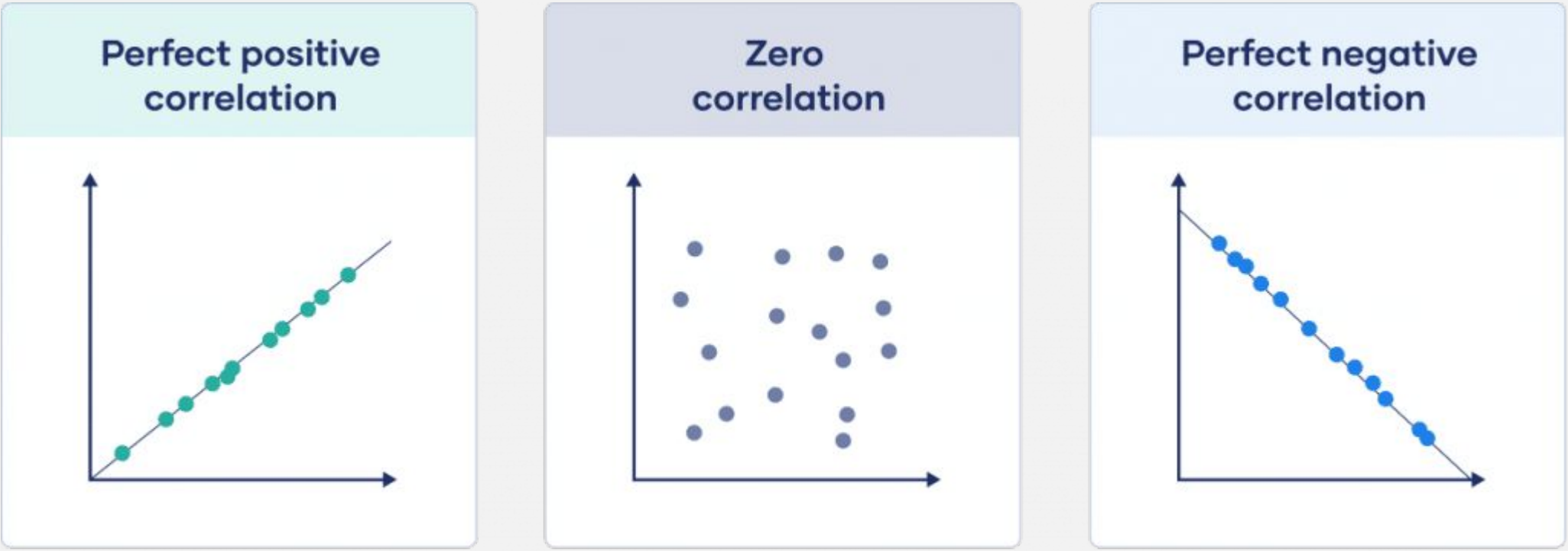
Category	Details
Definition	Correlation measures the **relationship between two variables** and how they move together
Purpose	To identify **strength and direction** of relationship between variables
Types of Correlation	**Positive** (both increase/decrease)**Negative** (one increases, other decreases)**Zero** (no relationship)
Correlation Coefficient (r)	Value ranges from **-1 to +1**
Formula	$r = \Sigma(X-\bar{X})(Y-\bar{Y}) \ / \ \sqrt{[\Sigma(X-\bar{X})^2 \ \Sigma(Y-\bar{Y})^2]}$
Strength Interpretation	+0.75 to +1 → Strong Positive+0.5 to +0.75 → Moderate0 to +0.5 → Weak0 → No Correlation-0.5 to 0 → Weak Negative-0.75 to -0.5 → Moderate Negative-1 to -0.75 → Strong Negative
Excel Functions	=CORREL(array1,array2)=PEARSON(array1,array2)
Input Requirement	Two datasets (X and Y values)
Output	Single value (r) showing relationship strength
Visualization Tool	Scatter Plot with Trendline
Advantages	Easy to calculateHelps in predictionUseful in analytics & ML
Limitations	Does not show cause-effectAffected by outliers
Applications	Sales vs AdsMarks vs Study HoursPrice vs Demand

Correlation Analysis



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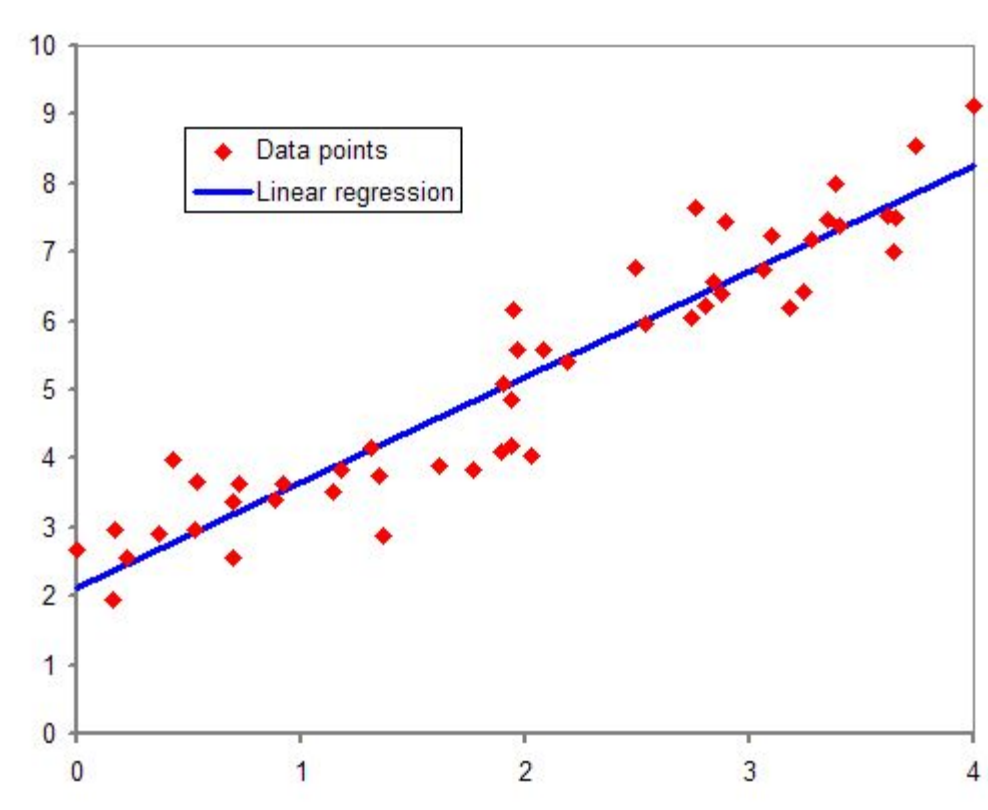
Regression Analysis

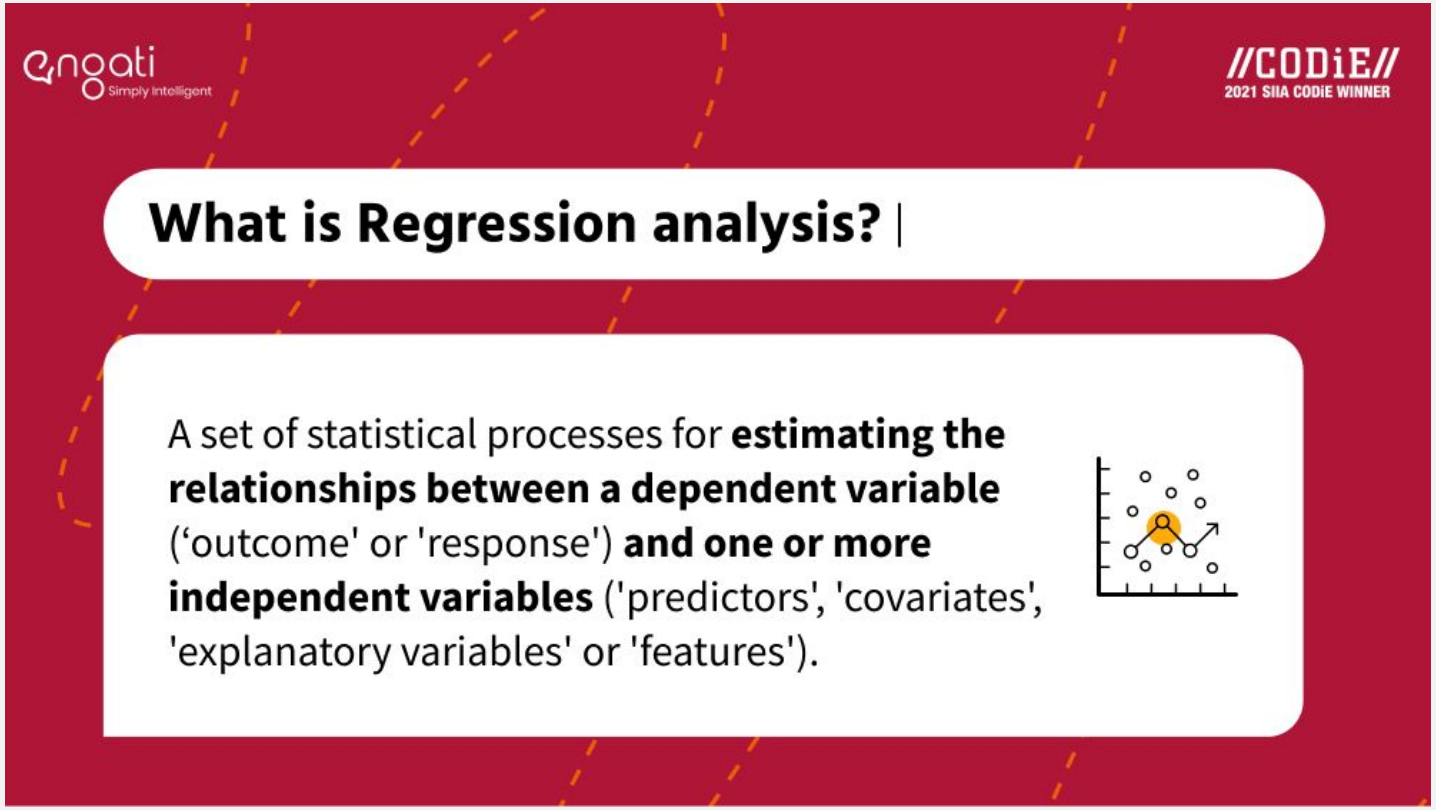
Category	Details
Definition	Regression Analysis is used to **predict the value of a dependent variable (Y)** based on one or more independent variables (X)
Purpose	To find **relationship and prediction** between variables
Types of Regression	**Simple Linear Regression** (1 independent variable)**Multiple Regression** (more than 1 independent variable)
Regression Equation	Y = a + bX
Terms Meaning	**Y** = Dependent Variable**X** = Independent Variable**a** = Intercept**b** = Slope
Excel Functions	=FORECAST.LINEAR(x, known_y's, known_x's)=LINEST(known_y's, known_x's)
Tool Used in Excel	**Data Analysis ToolPak → Regression**
Input Requirement	Y range (dependent)X range (independent)

Regression Analysis

Category	Details
Output	Coefficients, R^2, Standard Error, p-value
R^2 (Coefficient of Determination)	Shows **model accuracy (0 to 1)** → closer to 1 = better fit
Visualization	Scatter Plot + Trendline (Linear)
Steps in Excel	1. Enable Data Analysis ToolPak2. Go to Data → Data Analysis → Regression3. Select Input Y & X4. Choose Output Range5. Click OK
Advantages	Helps in predictionEasy to implementWidely used in analytics
Limitations	Assumes linear relationshipAffected by outliers
Applications	Sales predictionDemand forecastingRisk analysis
Output	Coefficients, R^2, Standard Error, p-value

Regression Analysis





Statistical Functions in Excel

Category	Function	Description	Syntax	Example
Central Tendency	AVERAGE()	Calculates mean (average)	=AVERAGE(range)	=AVERAGE(A1:A5)
	MEDIAN()	Finds middle value	=MEDIAN(range)	=MEDIAN(A1:A5)
	MODE.SNGL()	Most frequent value	=MODE.SNGL(range)	=MODE.SNGL(A1:A5)
Dispersion	MAX()	Maximum value	=MAX(range)	=MAX(A1:A5)
	MIN()	Minimum value	=MIN(range)	=MIN(A1:A5)
	VAR.S()	Sample variance	=VAR.S(range)	=VAR.S(A1:A5)
	VAR.P()	Population variance	=VAR.P(range)	=VAR.P(A1:A5)
	STDEV.S()	Sample standard deviation	=STDEV.S(range)	=STDEV.S(A1:A5)
	STDEV.P()	Population standard	=STDEV.P(range)	=STDEV.P(A1:A5)

Statistical Functions in Excel

Category	Function	Description	Syntax	Example
Counting Functions	COUNT()	Counts numeric values	=COUNT(range)	=COUNT(A1:A5)
	COUNTA()	Counts non-empty cells	=COUNTA(range)	=COUNTA(A1:A5)
	COUNTBLANK()	Counts empty cells	=COUNTBLANK(range)	=COUNTBLANK(A1:A5)
	COUNTIF()	Counts with condition	=COUNTIF(range,criteria)	=COUNTIF(A1:A5,">10")
Conditional Functions	SUMIF()	Sum with condition	=SUMIF(range,criteria)	=SUMIF(A1:A5,">10")
	AVERAGEIF()	Average with condition	=AVERAGEIF(range,criteria)	=AVERAGEIF(A1:A5,">10")

Statistical Functions in Excel

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Category	Function	Description	Syntax	Example
Ranking & Percentile	RANK()	Rank values	=RANK(number,range)	=RANK(A1,A1:A5)
	PERCENTILE()	Finds percentile	=PERCENTILE(range,k)	=PERCENTILE(A1:A5,0.5)
	QUARTILE()	Finds quartiles	=QUARTILE.INC(range,quart)	=QUARTILE.INC(A1:A5,1)
Correlation & Regression	CORREL()	Correlation coefficient	=CORREL(array1,array2)	=CORREL(A1:A5,B1:B5)
	FORECAST.LINEAR()	Predict future value	=FORECAST.LINEAR(x,known_y,known_x)	=FORECAST.LINEAR(10,B1:B5,A1:A5)
	LINEST()	Linear regression stats	=LINEST(y,x)	=LINEST(B1:B5,A1:A5)
Other Statistical Functions	GEOMEAN()	Geometric mean	=GEOMEAN(range)	=GEOMEAN(A1:A5)
	HARMEAN()	Harmonic mean	=HARMEAN(range)	=HARMEAN(A1:A5)
	AVEDEV()	Mean absolute deviation	=AVEDEV(range)	=AVEDEV(A1:A5)

Statistical Functions in Excel

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AVEDEV	average of absolute deviations of data elements from their mean
AVERAGE	average (i.e. mean) of a data set
AVERAGEIF	average of data elements satisfying some criteria (as for SUMIF)
AVERAGEIFS	average of data elements satisfying a set of criteria (as for SUMIFS)
COUNT	number of elements in a data set
DEVSQ	sum of squares of deviations
GEOMEAN	geometric mean of a data set
HARMEAN	harmonic mean of a data set
KURT	kurtosis of a data set
LARGE	kth largest element in a data set
MAX	largest element in a data set
MEDIAN	median of a data set
MIN	smallest element in a data set
MODE	mode of a data set
PERCENTRANK	percentage rank of an element in a data set
PERCENTILE	kth percentile of a data set
QUARTILE	kth quartile of a data set
RANK	rank of an element in a data set
SKEW	skewness of a data set
SMALL	kth smallest element in a data set
STANDARDIZE	standardized value of a data element
STDEV	sample standard deviation of a data set
STDEVP	population standard deviation of a data set
TRIMMEAN	mean of a data set excluding the smallest and largest elements
VAR	sample variance of a data set
VARP	population variance of a data set

CONFIDENCE.NORM	Equivalent to CONFIDENCE; based on normal distribution
CONFIDENCE.T	Confidence interval for Student's t distribution
COVARIANCE.P	Equivalent to COVAR, i.e. population covariance
COVARIANCE.S	Sample covariance
MODE.SNGL	Equivalent to MODE; single value
MODE.MULT	Array with all modes for a given range
PERCENTILE.INC	Equivalent to PERCENTILE; value p with $0 \le p \le 1$
PERCENTILE.EXC	Value p with $0 < p < 1$
PERCENTRANK.INC	Equivalent to PERCENTRANK; value p with $0 \le p \le 100$
PERCENTRANK.EXC	Value p with $0 < p < 100$
QUARTILE.INC	Equivalent to QUARTILE; value p with $0 \le p \le 1$
QUARTILE.EXC	Value p with $0 < p < 1$
RANK.EQ	Equivalent to RANK
RANK.AVG	Returns average rank if multiple data elements have the same rank
STDEV.P	Equivalent to STDEVP
STDEV.S	Equivalent to STDEV
VAR.P	Equivalent to VARP
VAR.S	Equivalent to VAR

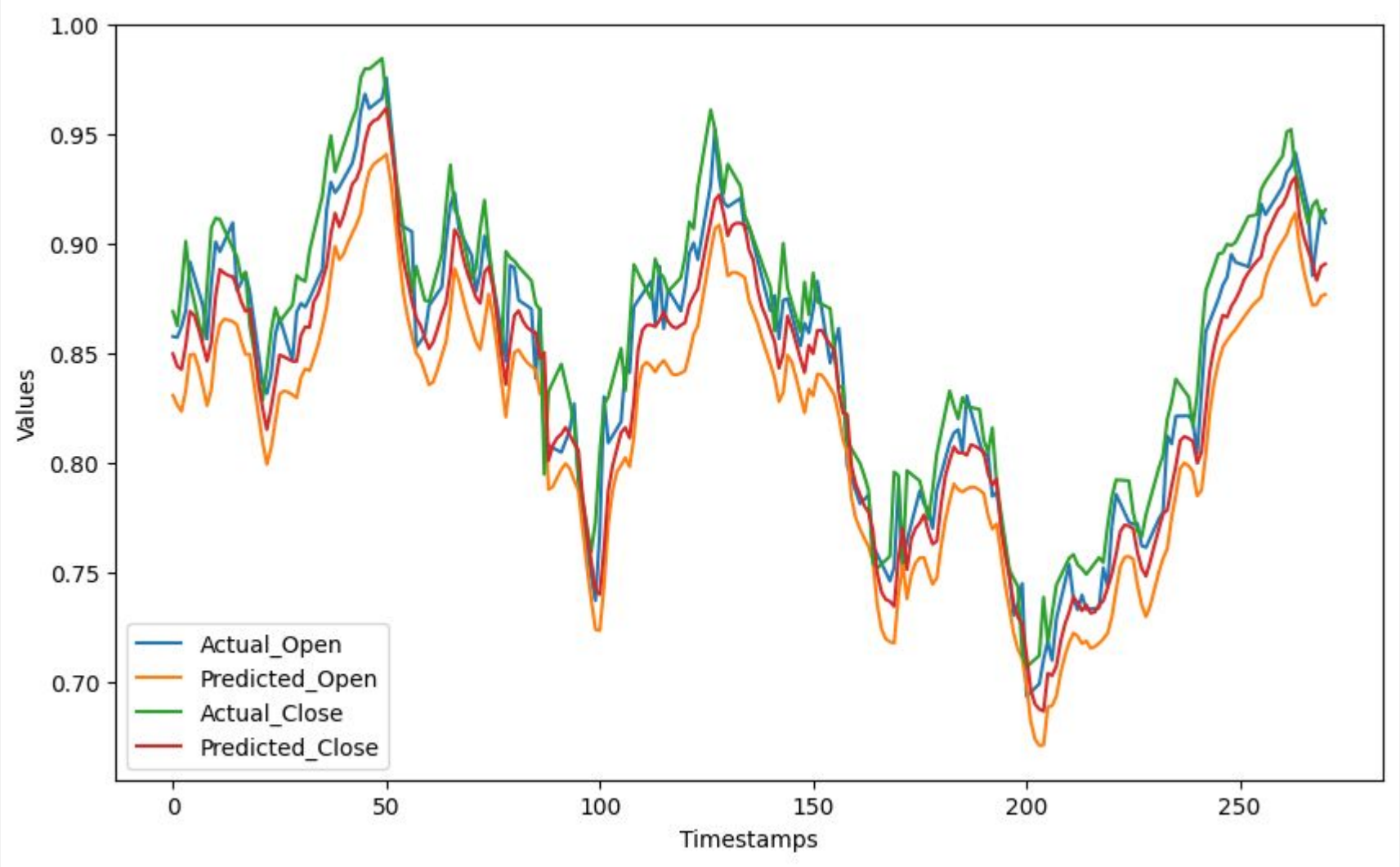
Time Series Analysis

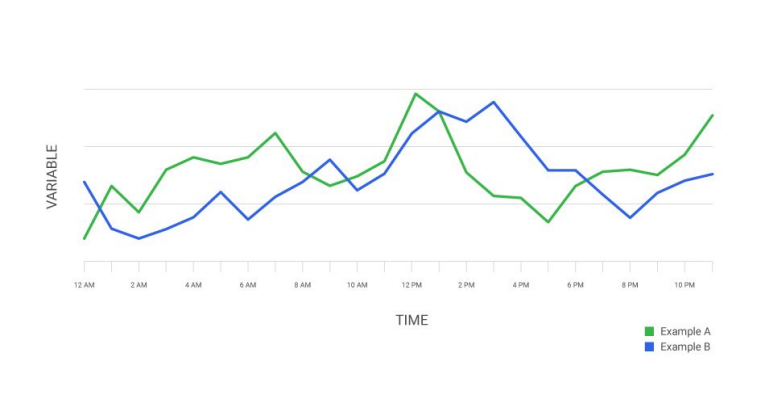
Category	Details
Definition	Time Series Analysis is used to analyze **data points collected over time** to identify patterns and trends
Purpose	To **forecast future values** based on past data
Components of Time Series	**Trend** (long-term movement)**Seasonality** (repeating patterns)**Cyclical** (long fluctuations)**Irregular** (random variations)
Types of Analysis	**Trend AnalysisSeasonal AnalysisForecasting**
Excel Functions	=FORECAST.LINEAR()=TREND()=GROWTH()
Advanced Excel Tool	**Forecast Sheet (Data → Forecast Sheet)**
Input Requirement	Time data (Date/Time)Values (Sales, Temperature, etc.)

Time Series Analysis

Category	Details
Output	Predicted future values, trend line, confidence interval
Visualization	Line Chart (best for time-based data)
Steps in Excel	1. Enter time & data2. Select data3. Insert → Line Chart4. Use Forecast Sheet or formulas
TREND Function	Predicts values using linear trend
GROWTH Function	Predicts exponential growth
Advantages	Helps in forecastingEasy visualizationUseful in business planning
Limitations	Assumes past patterns continueAffected by sudden changes
Applications	Sales forecastingStock market analysisWeather prediction

Time Series Analysis





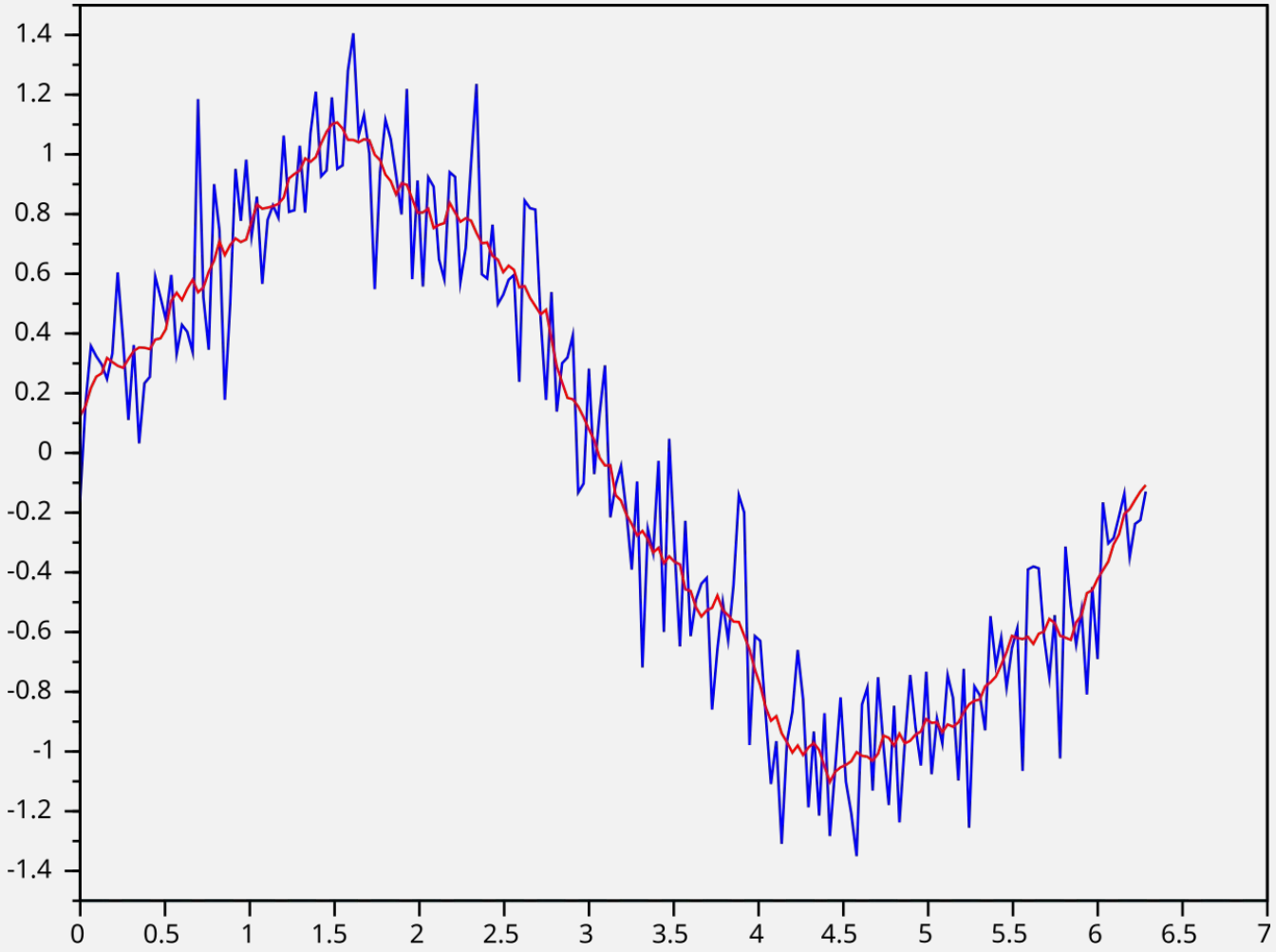
Moving Average

Category	Details
Definition	Moving Average is used to **smooth time series data** by calculating averages of different subsets of data
Purpose	To **identify trends** and reduce short-term fluctuations
Types of Moving Average	**Simple Moving Average (SMA)Weighted Moving Average (WMA)Exponential Moving Average (EMA)**
Formula (SMA)	Moving Average = (Sum of n values) / n
Excel Method (Manual)	Use =AVERAGE(range) for each window
Excel Tool	Data Analysis ToolPak → Moving Average
Input Requirement	Time-based data (e.g., daily, monthly values)

Moving Average

Category	Details
Output	Smoothed values showing trend
Visualization	Line Chart with moving average trendline
Steps in Excel (Manual)	1. Enter data2. Select window (e.g., 3-period)3. Apply =AVERAGE(A1:A3)4. Drag formula down
Steps (ToolPak)	1. Go to Data → Data Analysis2. Select Moving Average3. Choose Input Range4. Enter Interval (e.g., 3)5. Click OK
Advantages	Smooths dataEasy to calculateHelps in forecasting
Limitations	Lags behind actual valuesNot suitable for sudden changes
Applications	Stock pricesSales trendsWeather data

Moving Average



Exponential Moving Average

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Category	Details
Definition	EMA is a type of moving average that gives **more weight to recent data points**
Purpose	To **track trends more quickly** than Simple Moving Average (SMA)
Key Feature	Recent values have **higher importance**, older values have less impact
Formula (EMA)	EMA = (Current Value × α) + (Previous EMA × (1 − α))
Smoothing Factor (α)	α = 2 / (n + 1), where n = number of periods
Excel Calculation	No direct function → calculated manually using formula
Input Requirement	Time series data (e.g., daily sales, stock prices)
Output	Smoothed trend line with more sensitivity to recent changes
Visualization	Line chart comparing actual data vs EMA
Steps in Excel	1. Enter data2. Calculate α3. First EMA = Simple Average4. Apply EMA formula5. Drag formula down
Advantages	Faster response to changesBetter for forecastingWidely used in finance
Limitations	Slightly complexStill affected by extreme values
Applications	Stock market analysisDemand forecastingSignal processing

Exponential Moving Average

Applications of Excel in Data Analytics

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Category	Application	Description	Excel Tools/Functions Used
Data Collection	Data Entry & Import	Collecting data from files, databases, APIs	Import Wizard, Power Query
Data Cleaning	Data Preprocessing	Removing errors, duplicates, handling missing values	TRIM(), CLEAN(), IF(), Remove Duplicates
Data Transformation	Data Formatting	Converting data into usable format	Text to Columns, UPPER(), LOWER(), PROPER()
Data Analysis	Statistical Analysis	Performing mean, median, variance, etc.	AVERAGE(), MEDIAN(), STDEV()
Data Summarization	Aggregation	Summarizing large datasets	Pivot Tables
Data Visualization	Charts & Graphs	Visual representation of data	Bar Chart, Line Chart, Pie Chart
Trend Analysis	Time Series Analysis	Identifying patterns over time	Line Charts, Moving Average
Forecasting	Prediction	Predicting future values	FORECAST.LINEAR(), Trendline

Applications of Excel in Data Analytics

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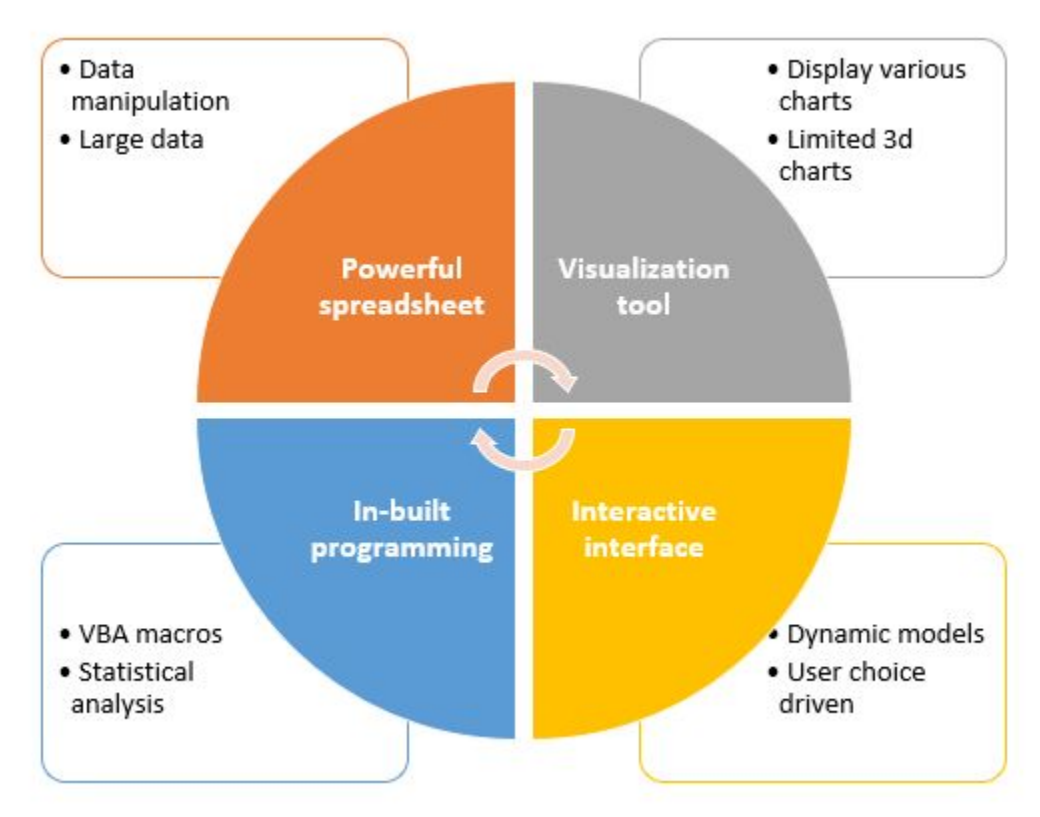
Category	Application	Description	Excel Tools/Functions Used
Correlation Analysis	Relationship Analysis	Finding relationship between variables	CORREL()
Regression Analysis	Predictive Modeling	Predicting dependent variable	Data Analysis ToolPak, LINEST()
What-if Analysis	Scenario Testing	Testing different scenarios	Goal Seek, Scenario Manager
Data Validation	Error Control	Restricting invalid inputs	Data Validation Tool
Dashboard Creation	Reporting	Creating interactive dashboards	Charts, Slicers, Pivot Charts
Automation	Repetitive Tasks	Automating processes	Macros, VBA
Business Analytics	Decision Making	Supporting business decisions	KPIs, Financial Models

Applications of Excel in Data Analytics



THANK YOU